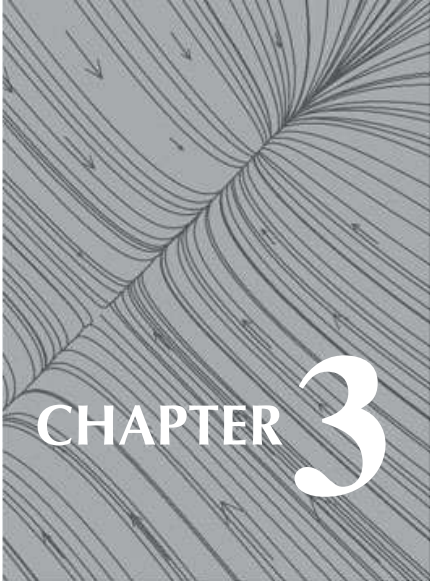




CHAPTER 3

Linear Systems

In bear country, wear bells to avoid surprising bruises. Don't approach a bear, but don't run away from one either. Movement may incite a charge, and a bear can outrun you.

If a bear approaches you, talk to it calmly in a low, firm voice. Slowly back away until the bear is out of sight. If you climb a tree, get at least 12 feet up.

If a bear senses that its food may be stolen or that its cub is in danger, it may charge. If this happens, drop to the ground and assume a fetal position, protecting your head with your arms, and play dead. Keep your pack on; it may serve as armor.

—*Reader's Digest Practical Problem Solver* [241]

We begin our discussion of linear system theory with a description of the concept of linearity by Mitchell J. Feigenbaum [227, p. 3], who is a researcher of nonlinear systems. “Linearity means that the rule that determines what a piece of a system is going to do next is not influenced by what it is doing now. More precisely, this is intended in a differential or incremental sense: For a linear spring, the increase of its tension is proportional to the increment whereby it is stretched, with the ratio of these increments exactly independent of how much it has already been stretched. Such a spring can be stretched arbitrarily far, and in particular will never snap or break. Accordingly, no real spring is linear.” The real world is nonlinear. However, there are a number of reasons to investigate linear systems. Linear models are often used to approximate nonlinear systems. Many times, this approximation is sufficient for the controller design for the underlying nonlinear system. Knowledge of linear system theory helps to understand the intricate theory of nonlinear systems.

In this chapter we acquaint ourselves with basic concepts from linear systems that will be used in subsequent chapters.

3.1 Reachability and Controllability

We first consider a linear, time-invariant, discrete-time system modeled by

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (3.1)$$

where $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$. The solution to (3.1) is

$$\mathbf{x}(i) = A^i \mathbf{x}(0) + \sum_{k=0}^{i-1} A^{i-k-1} B \mathbf{u}(k),$$

which can be represented as

$$\mathbf{x}(i) = A^i \mathbf{x}(0) + [B \quad AB \quad \dots \quad A^{i-1}B] \begin{bmatrix} \mathbf{u}(i-1) \\ \mathbf{u}(i-2) \\ \vdots \\ \mathbf{u}(0) \end{bmatrix}. \quad (3.2)$$

Let

$$U_i = [B \quad AB \quad \dots \quad A^{i-1}B], \quad i = 1, 2, \dots$$

Note that $U_i \in \mathbb{R}^{n \times mi}$. If for some positive l we have

$$\text{rank } U_l = \text{rank } U_{l+1},$$

then all the columns of $A^l B$ are linearly dependent on those of U_l . This, in turn, implies that all the columns of $A^{l+1}B, A^{l+2}B, \dots$ must also be linearly dependent on those of U_l . Hence,

$$\text{rank } U_l = \text{rank } U_{l+1} = \text{rank } U_{l+2} = \dots \quad (3.3)$$

The above implies that the rank of U_i increases by at least one when i is increased by one, until the maximum value of rank U_i is attained. The maximum value of the rank U_i is guaranteed to be achieved for $i = n$, which means that we always have

$$\text{rank } U_n = \text{rank } U_{n+1} = \text{rank } U_{n+2} = \dots \quad (3.4)$$

It follows from (3.3) that if rank $B = p$, then

$$\text{rank}[B \quad AB \quad \dots \quad A^{n-1}B] = \text{rank}[B \quad AB \quad \dots \quad A^{n-p}B]. \quad (3.5)$$

We say that the system (3.1) is **reachable** if for any $\mathbf{x}_f$ there exists an integer $q > 0$ and a control sequence, $\{\mathbf{u}(i) : i = 0, 1, \dots, q-1\}$, that transfers $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(q) = \mathbf{x}_f$.

Theorem 3.1 The system (3.1) is reachable if and only if

$$\text{rank}[B \quad AB \quad \dots \quad A^{n-1}B] = n.$$

Proof Using (3.2) where $\mathbf{x}(0) = \mathbf{0}$ and $\mathbf{x}(q) = \mathbf{x}_f$, we obtain

$$\begin{aligned} \mathbf{x}_f &= \sum_{k=0}^{q-1} \mathbf{A}^{q-k-1} \mathbf{B} \mathbf{u}(k) \\ &= [\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{q-1} \mathbf{B}] \begin{bmatrix} \mathbf{u}(q-1) \\ \mathbf{u}(q-2) \\ \vdots \\ \mathbf{u}(0) \end{bmatrix}. \end{aligned} \quad (3.6)$$

It follows from the above that for any $\mathbf{x}_f$ there exists a control sequence $\{\mathbf{u}(i) : i = 0, 1, \dots, q-1\}$ that transfers $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(q) = \mathbf{x}_f$ if and only if

$$\text{rank}[\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{q-1} \mathbf{B}] = \text{rank } \mathbf{U}_q = n.$$

By (3.4) the maximum value of the rank of $\mathbf{U}_q$ is guaranteed to be attained for $q = n$, which proves the theorem.

A notion closely related to reachability is the notion of *controllability*. The notion of controllability is related to the property of a controlled system being transferable from any given state to the origin $\mathbf{0}$ of the state space by means of an appropriately chosen control law.

Theorem 3.2 The system (3.1) is controllable if and only if

$$\text{range}(\mathbf{A}^n) \subset \text{range}[\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{n-1} \mathbf{B}].$$

Proof Using the solution formula for the system (3.1), for the case when $\mathbf{x}(0) = \mathbf{x}_0$ and $\mathbf{x}_f = \mathbf{x}(q) = \mathbf{0}$, we obtain

$$\begin{aligned} \mathbf{A}^q \mathbf{x}_0 &= - \sum_{k=0}^{q-1} \mathbf{A}^{q-k-1} \mathbf{B} \mathbf{u}(k) \\ &= -[\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{q-1} \mathbf{B}] \begin{bmatrix} \mathbf{u}(q-1) \\ \mathbf{u}(q-2) \\ \vdots \\ \mathbf{u}(0) \end{bmatrix}. \end{aligned} \quad (3.7)$$

The system (3.1) is controllable if and only if for some $q > 0$ and arbitrary initial condition $\mathbf{x}_0$, the vector $\mathbf{A}^q \mathbf{x}_0$ is in the range of $[\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{q-1} \mathbf{B}] = \mathbf{U}_q$. Taking into account that the maximal range of $\mathbf{U}_q$ is guaranteed to be attained for $q = n$, the system (3.1) is controllable if and only if the condition of the theorem is satisfied.

Note that the condition of the above theorem can be equivalently represented as

$$\text{rank}[\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{n-1} \mathbf{B}] = \text{rank}[\mathbf{B} \quad \mathbf{AB} \quad \dots \quad \mathbf{A}^{n-1} \mathbf{B} \quad \mathbf{A}^n]. \quad (3.8)$$

We illustrate the above discussion with a simple example that we found in Kaczorek [144, p. 482].

◆ Example 3.1

Consider a discrete-time system modeled by

$$\begin{aligned}\mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{b}u(k) \\ &= \begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(k), \quad a \in \mathbb{R}.\end{aligned}\quad (3.9)$$

The above system model is not reachable because

$$\text{rank}[\mathbf{b} \quad \mathbf{A}\mathbf{b}] = \text{rank} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = 1 < 2.$$

Using (3.6), it is easy to check that, for example, the state $\mathbf{x}_f = [0 \ 1]^T$ is not reachable from $\mathbf{x}(0) = \mathbf{0}$ because $\mathbf{x}_f$ does not belong to the range of $[\mathbf{b} \quad \mathbf{A}\mathbf{b}]$. In fact, nonreachable states form the set

$$\left\{ c \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad c \in \mathbb{R} \right\}.$$

However, the system (3.9) is controllable because for any $a \in \mathbb{R}$,

$$\text{rank}[\mathbf{b} \quad \mathbf{A}\mathbf{b}] = \text{rank} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \text{rank}[\mathbf{b} \quad \mathbf{A}\mathbf{b} \quad \mathbf{A}^2\mathbf{b}] = \text{rank} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix};$$

that is,

$$\text{range}(\mathbf{A}^2) = \text{range}(\mathbf{O}) \subset \text{range}[\mathbf{b} \quad \mathbf{A}\mathbf{b}].$$

(See appendix, Section A.3, for an introduction to the $\mathbf{O}$ notation.) Using (3.7), we can easily see that arbitrary initial state $\mathbf{x}(0) \in \mathbb{R}^2$ of the system (3.9) can be transferred to the origin of $\mathbb{R}^2$ using, for example, the zero control sequence, $u(0) = u(1) = 0$. This is because $\mathbf{A}^2 = \mathbf{O}$ and hence $\mathbf{A}^2\mathbf{x}(0) = \mathbf{0}$ for any $\mathbf{x}(0) \in \mathbb{R}^2$.

In summary, for discrete-time linear systems, reachability implies controllability, and the two notions are equivalent if the matrix $\mathbf{A}$ of the given discrete system is nonsingular.

We next discuss the notions of reachability and controllability for continuous-time linear systems.

We say that the system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$, where $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times m}$, or equivalently the pair $(\mathbf{A}, \mathbf{B})$, is reachable if for any $\mathbf{x}_f \in \mathbb{R}^n$ there is t_1 and a control law $\mathbf{u}(t)$ that transfers $\mathbf{x}(t_0) = \mathbf{0}$ to $\mathbf{x}(t_1) = \mathbf{x}_f$.

We say that the system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$, or equivalently the pair $(\mathbf{A}, \mathbf{B})$, is *controllable* if there is a control law $\mathbf{u}(t)$ that transfers any initial state $\mathbf{x}_0 = \mathbf{x}(t_0)$ at any time t_0 to the origin $\mathbf{0}$ of the state space at some time $t_1 > t_0$.

It turns out that in the case of continuous, time-invariant systems the two notions are equivalent. Indeed, from the solution formula for the controlled system (A.151), for $\mathbf{x}(0) = \mathbf{0}$ and

$\mathbf{x}(t_1) = \mathbf{x}_f$, we obtain

$$\mathbf{x}_f = \int_0^{t_1} e^{A(t_1-t)} \mathbf{B} \mathbf{u}(t) dt.$$

Premultiplying the above equation by e^{-At_1} and rearranging, we get

$$\mathbf{v} = \int_0^{t_1} e^{-At} \mathbf{B} \mathbf{u}(t) dt, \quad (3.10)$$

where $\mathbf{v} = e^{-At_1} \mathbf{x}_f$. On the other hand, from the solution formula for the controlled system (A.151) for $\mathbf{x}(0) = \mathbf{x}_0$ and $\mathbf{x}(t_1) = \mathbf{0}$, we obtain

$$\mathbf{0} = e^{At_1} \mathbf{x}_0 + \int_0^{t_1} e^{A(t_1-t)} \mathbf{B} \mathbf{u}(t) dt.$$

Premultiplying the above equation by e^{-At_1} and rearranging yields

$$-\mathbf{x}_0 = \int_0^{t_1} e^{-At} \mathbf{B} \mathbf{u}(t) dt. \quad (3.11)$$

Comparing (3.10) and (3.11), we conclude that reachability of the system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$ is equivalent to its controllability. This is not true in the case of discrete-time systems, where reachability implies controllability, and the two notions are equivalent if the matrix $\mathbf{A}$ of the given discrete system is nonsingular.

There are many criteria for checking if a given system is reachable/controllable or not. We present a few of them in this section.

Theorem 3.3 The following are equivalent:

1. The system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$ is reachable.
2. $\text{rank}[\mathbf{B} \ \mathbf{AB} \ \cdots \ \mathbf{A}^{n-1}\mathbf{B}] = n$.
3. The n rows of the matrix $e^{-At} \mathbf{B}$ are linearly independent over the set of real numbers for all $t \in [0, \infty)$.
4. The matrix

$$\mathbf{W}(t_0, t_1) = \int_{t_0}^{t_1} e^{-At} \mathbf{B} \mathbf{B}^T e^{-A^T t} dt$$

is nonsingular for all $t_1 > t_0$.

Proof We first prove the implication $1 \Rightarrow 2$; that is, if the pair $(\mathbf{A}, \mathbf{B})$ is reachable, then the so-called controllability matrix

$$[\mathbf{B} \ \mathbf{AB} \ \cdots \ \mathbf{A}^{n-1}\mathbf{B}] \in \mathbb{R}^{n \times mn}$$

is of full rank n . We prove this statement by contraposition. Thus, we assume that

$$\text{rank}[\mathbf{B} \ \mathbf{AB} \ \cdots \ \mathbf{A}^{n-1}\mathbf{B}] < n;$$

then, there is a constant n -vector $\mathbf{q} \neq \mathbf{0}$ so that

$$\mathbf{q}^T \mathbf{B} = \mathbf{0}^T, \quad \mathbf{q}^T \mathbf{A} \mathbf{B} = \mathbf{0}^T, \dots, \mathbf{q}^T \mathbf{A}^{n-1} \mathbf{B} = \mathbf{0}^T.$$

By the Cayley–Hamilton theorem the matrix $\mathbf{A}$ satisfies its own characteristic equation. Hence

$$\mathbf{A}^n = -a_{n-1} \mathbf{A}^{n-1} - \dots - a_1 \mathbf{A} - a_0 \mathbf{I}_n.$$

Therefore

$$\mathbf{q}^T \mathbf{A}^n \mathbf{B} = \mathbf{q}^T (-a_{n-1} \mathbf{A}^{n-1} \mathbf{B} - \dots - a_1 \mathbf{A} \mathbf{B} - a_0 \mathbf{B}) = \mathbf{0}^T.$$

By induction we obtain

$$\mathbf{q}^T \mathbf{A}^i \mathbf{B} = \mathbf{0}^T \quad \text{for } i = n+1, n+2, \dots$$

Now let $\mathbf{x}(0) = \mathbf{0}$ and $\mathbf{x}(t_1) = \mathbf{q}$. We will now show that there is no control law that can transfer the system from $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(t_1) = \mathbf{q}$. From the solution formula for the controlled system, we obtain

$$\mathbf{q} = \int_0^{t_1} e^{\mathbf{A}(t_1-t)} \mathbf{B} \mathbf{u}(t) dt.$$

We now premultiply the resulting equation by $\mathbf{q}^T$ to get

$$0 \neq \|\mathbf{q}\|^2 = \int_0^{t_1} \mathbf{q}^T e^{\mathbf{A}(t_1-t)} \mathbf{B} \mathbf{u}(t) dt = 0$$

because

$$\mathbf{q}^T e^{\mathbf{A}(t_1-t)} \mathbf{B} = \mathbf{q}^T \left(\mathbf{B} + (t_1 - t) \mathbf{A} \mathbf{B} + \frac{(t_1 - t)^2}{2!} \mathbf{A}^2 \mathbf{B} - \dots \right) = \mathbf{0}^T.$$

Thus the system cannot be transferred from $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(t_1) = \mathbf{q}$, and hence the pair $(\mathbf{A}, \mathbf{B})$ is not reachable.

We will now prove the implication $2 \Rightarrow 3$; that is, if the controllability matrix is of full rank, then the n rows of $e^{-\mathbf{A}t} \mathbf{B}$ are linearly independent over $\mathbb{R}$ for all $t \in [0, \infty)$. We prove this statement by contraposition. So assume that for some nonzero n -vector $\mathbf{q}$ we have

$$\mathbf{q}^T e^{-\mathbf{A}t} \mathbf{B} = \mathbf{0}^T, \quad \text{for all } t \in [0, \infty).$$

If we set $t = 0$, then we obtain

$$\mathbf{q}^T \mathbf{B} = \mathbf{0}^T.$$

Differentiating $\mathbf{q}^T e^{-\mathbf{A}t} \mathbf{B} = \mathbf{0}$ and letting $t = 0$ yields

$$\mathbf{q}^T \mathbf{A} \mathbf{B} = \mathbf{0}^T.$$

Continuing this process gives

$$\mathbf{q}^T [\mathbf{B} \quad \mathbf{A} \mathbf{B} \quad \dots \quad \mathbf{A}^{n-1} \mathbf{B}] = \mathbf{0}^T,$$

which means that $\text{rank}[\mathbf{B} \quad \mathbf{A} \mathbf{B} \quad \dots \quad \mathbf{A}^{n-1} \mathbf{B}] < n$.

We will now prove, by contraposition, the implication $3 \Rightarrow 4$; that is, if the n rows of $e^{-\mathbf{A}t} \mathbf{B}$ are linearly independent over $\mathbb{R}$ for all $t \in [0, \infty)$, then $\det \mathbf{W}(t_0, t_1) \neq 0$ for any $t_1 > t_0$. So assume that $\det \mathbf{W}(t_0, t_1) = 0$. Then, there is a nonzero n -vector

$\mathbf{q} \in \mathbb{R}^n$ such that

$$\mathbf{q}^T \int_{t_0}^{t_1} e^{-\mathbf{A}t} \mathbf{B} \mathbf{B}^T e^{-\mathbf{A}^T t} dt = \mathbf{0}^T$$

for all $t \in [t_0, t_1]$. We postmultiply the above equation by $\mathbf{q}$ to get

$$\int_{t_0}^{t_1} (\mathbf{q}^T e^{-\mathbf{A}t} \mathbf{B})(\mathbf{B}^T e^{-\mathbf{A}^T t} \mathbf{q}) dt = 0.$$

Let $\mathbf{v}(t) = \mathbf{B}^T e^{-\mathbf{A}^T t} \mathbf{q}$; then the above equation can be written as

$$\int_{t_0}^{t_1} \|\mathbf{v}(t)\|^2 dt = 0.$$

However, the above is true if and only if $\mathbf{v}(t) = \mathbf{0}$ —that is, if and only if the rows of $e^{-\mathbf{A}t} \mathbf{B}$ are linearly dependent over the set of real numbers.

We now prove, constructively, the implication $4 \Rightarrow 1$. Let $\mathbf{x}(t_0) = \mathbf{x}_0$ be an arbitrary initial state and let $\mathbf{x}(t_1) = \mathbf{x}_f$ be an arbitrary final state. Then, we claim that the control law

$$\mathbf{u}(t) = \mathbf{B}^T e^{-\mathbf{A}^T t} \mathbf{W}^{-1}(t_0, t_1)(e^{-\mathbf{A}t_1} \mathbf{x}(t_1) - e^{-\mathbf{A}t_0} \mathbf{x}_0) \quad (3.12)$$

will transfer the system from $\mathbf{x}_0$ to $\mathbf{x}_f$. Note that to construct the above control law the matrix $\mathbf{W}(t_0, t_1)$ must be invertible. Substituting (3.12) into

$$\mathbf{x}(t_1) = e^{\mathbf{A}(t_1-t_0)} \mathbf{x}_0 + \int_{t_0}^{t_1} e^{\mathbf{A}(t_1-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau$$

yields

$$\begin{aligned} \mathbf{x}(t_1) &= e^{\mathbf{A}(t_1-t_0)} \mathbf{x}_0 \\ &+ e^{\mathbf{A}t_1} \int_{t_0}^{t_1} e^{-\mathbf{A}\tau} \mathbf{B} \mathbf{B}^T e^{-\mathbf{A}^T \tau} d\tau \mathbf{W}^{-1}(t_0, t_1)(e^{-\mathbf{A}t_1} \mathbf{x}(t_1) - e^{-\mathbf{A}t_0} \mathbf{x}_0). \end{aligned}$$

Performing easy manipulations gives $\mathbf{x}(t_1) = \mathbf{x}(t_1)$, which means that indeed the proposed control law does the required job. Thus, we have established equivalence between the four statements of the theorem.

◆ Example 3.2

Consider an armature-controlled DC motor system whose schematic is shown in Figure 3.1, where the system parameters are $R_a = 1 \Omega$, $L_a \approx 0 \text{ H}$, $K_b = 5 \text{ V/rad/sec}$, $K_j = 5 \text{ Nm/A}$, and the effective moment of inertia is $I_{eq} = 0.1 \text{ kg}\cdot\text{m}^2$. The gear inertia and friction are negligible.

Applying Kirchhoff's voltage law to the armature circuit gives

$$R_a i_a + K_b \dot{\theta} = e_a. \quad (3.13)$$

The armature-controlled DC motor uses a constant field current, and therefore the motor torque is

$$\tau_m = K_j i_a.$$

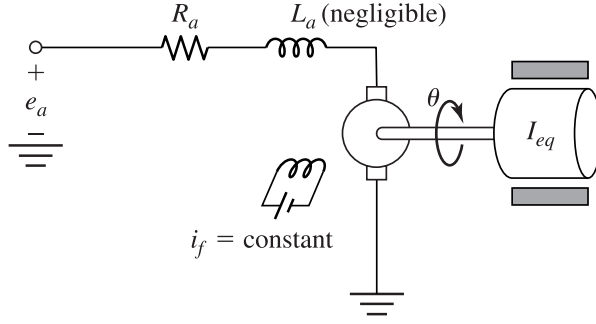


Figure 3.1 Schematic of an armature-controlled DC motor system of Example 3.2.

The motor torque is equal to the torque delivered to the load. Hence,

$$I_{eq} \ddot{\theta} = K_i i_a. \quad (3.14)$$

Substituting i_a from (3.13) into the above equation and dividing both sides by I_{eq} yields

$$\begin{aligned} \ddot{\theta} &= \frac{K_i}{I_{eq}} \left(\frac{e_a - K_b \dot{\theta}}{R_a} \right) \\ &= -\frac{K_i K_b}{I_{eq} R_a} \dot{\theta} + \frac{K_i}{I_{eq} R_a} e_a. \end{aligned}$$

Let $x_1 = \theta$, $x_2 = \dot{\theta}$, and $u = e_a$. Then taking into account the system parameters, we represent the above equation in a state-space format:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -250 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 50 \end{bmatrix} u.$$

We will now find a control u that transfers the system from the initial state $\mathbf{x}(t_0) = \mathbf{x}(0) = [10 \ 1]^T$ to the final state $\mathbf{x}(2) = [0 \ 0]^T$. Recall that the solution of a controlled system has the form

$$\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)} \mathbf{x}(t_0) + \int_{t_0}^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau.$$

Our goal is to find u so that

$$\mathbf{0} = e^{\mathbf{A}2} \mathbf{x}(0) + \int_0^2 e^{\mathbf{A}(2-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau.$$

One can verify that one such u has the form

$$u(t) = -\mathbf{B}^T e^{\mathbf{A}^T(2-t)} \left(\int_0^2 e^{\mathbf{A}(2-\tau)} \mathbf{B} \mathbf{B}^T e^{\mathbf{A}^T(2-\tau)} d\tau \right)^{-1} e^{\mathbf{A}2} \mathbf{x}(0),$$

where

$$e^{\mathbf{A}t} = \begin{bmatrix} 1 & \frac{1}{250}(1 - e^{-250t}) \\ 0 & e^{-250t} \end{bmatrix}.$$

Hence,

$$\begin{aligned} \int_0^2 e^{\mathbf{A}(2-\tau)} \mathbf{B} \mathbf{B}^T e^{\mathbf{A}^T(2-\tau)} d\tau &= \int_0^2 \begin{bmatrix} 1 & \frac{1}{250} (1 - e^{-250(2-\tau)}) \\ 0 & e^{-250(2-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 50 \end{bmatrix} [0 \ 50] \\ &\quad \times \begin{bmatrix} 1 & 0 \\ \frac{1}{250} (1 - e^{-250(2-\tau)}) & e^{-250(2-\tau)} \end{bmatrix} d\tau \\ &= \begin{bmatrix} 0.07976 & 0.02 \\ 0.02 & 5 \end{bmatrix}. \end{aligned}$$

The inverse of the above matrix is

$$\begin{bmatrix} 12.5502 & -0.0502 \\ -0.0502 & 0.2002 \end{bmatrix}.$$

Hence,

$$\begin{aligned} u(t) &= -[0 \ 50] \begin{bmatrix} 1 & 0 \\ \frac{1}{250} (1 - e^{-250(2-t)}) & e^{-250(2-t)} \end{bmatrix} \\ &\quad \times \begin{bmatrix} 12.5502 & -0.0502 \\ -0.0502 & 0.2002 \end{bmatrix} e^{2\mathbf{A}} \begin{bmatrix} 10 \\ 1 \end{bmatrix} \\ &= 50.22e^{(250t-500)} - 25.11. \end{aligned}$$

The above control strategy transfers the system from $\mathbf{x}(0) = [10 \ 1]^T$ to $\mathbf{x}(2) = [0 \ 0]^T$. Plots of x_1 , x_2 versus time are shown in Figure 3.2. A plot of u versus time is given in Figure 3.3.

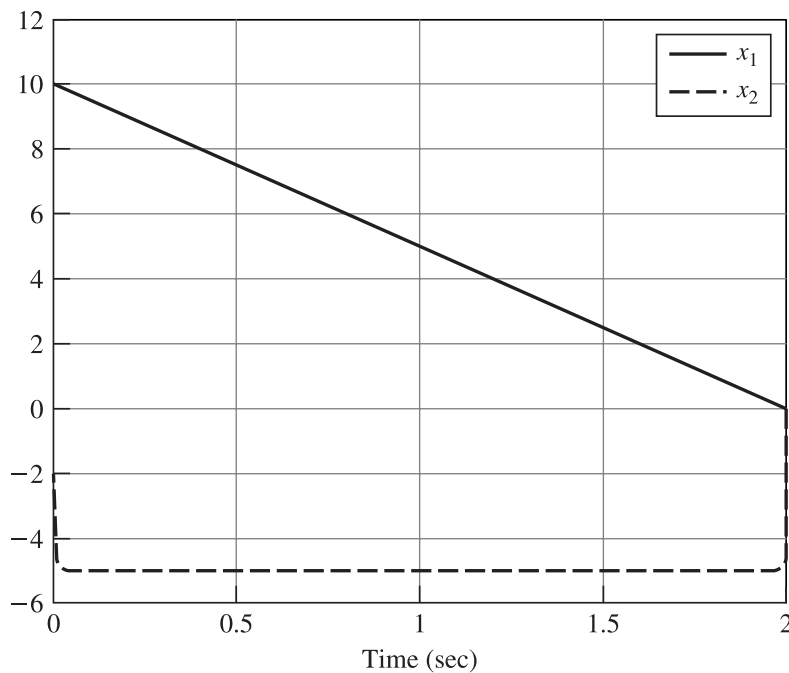


Figure 3.2 Plots of state variables versus time of Example 3.2.

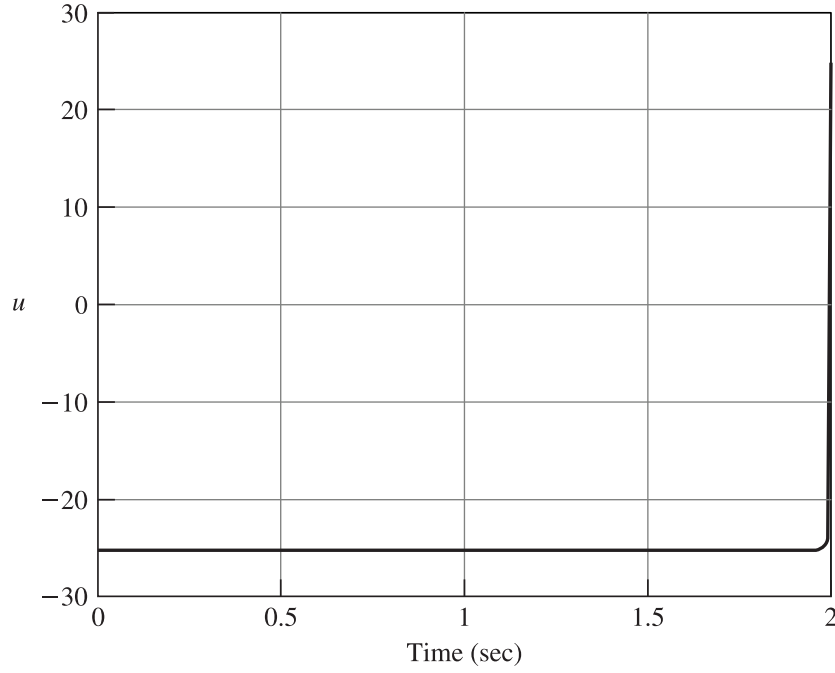


Figure 3.3 A plot of the control action versus time of Example 3.2.

We now give a necessary and sufficient condition for the pair $(\mathbf{A}, \mathbf{B})$ to be nonreachable.

Theorem 3.4 The pair $(\mathbf{A}, \mathbf{B})$ is nonreachable if and only if there is a similarity transformation $\mathbf{z} = \mathbf{T}\mathbf{x}$ such that

$$\tilde{\mathbf{A}} = \mathbf{T}\mathbf{A}\mathbf{T}^{-1} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{O} & \mathbf{A}_4 \end{bmatrix}, \quad \tilde{\mathbf{B}} = \mathbf{T}\mathbf{B} = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{O} \end{bmatrix},$$

where the pair $(\mathbf{A}_1, \mathbf{B}_1)$ is reachable, $\mathbf{A}_1 \in \mathbb{R}^{r \times r}$, $\mathbf{B}_1 \in \mathbb{R}^{r \times m}$, and the rank of the controllability matrix of the pair $(\mathbf{A}, \mathbf{B})$ equals r .

Proof We first prove necessity ($\Rightarrow$). After forming the controllability matrix of the pair $(\mathbf{A}, \mathbf{B})$, we use elementary row operations to get

$$\begin{aligned} \mathbf{T}[\mathbf{B} \quad \mathbf{AB} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}] &= [\mathbf{TB} \quad (\mathbf{TAT}^{-1})(\mathbf{TB}) \quad \cdots \quad (\mathbf{TAT}^{-1})^{n-1}(\mathbf{TB})] \\ &= \begin{bmatrix} x & x & x & \cdots & x & x & x \\ 0 & x & x & \cdots & x & x & x \\ \vdots & & & \vdots & & & \vdots \\ 0 & 0 & 0 & \cdots & x & x & x \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & & & \vdots & & & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{B}_1 & \mathbf{A}_1\mathbf{B}_1 & \mathbf{A}_1^2\mathbf{B}_1 & \cdots & \mathbf{A}_1^{n-1}\mathbf{B}_1 \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \cdots & \mathbf{O} \end{bmatrix}, \end{aligned} \quad (3.15)$$

where the symbol x denotes a “don’t care”—that is, an unspecified scalar. We now show that indeed

$$\tilde{\mathbf{A}} = \mathbf{T} \mathbf{A} \mathbf{T}^{-1} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{A}_3 & \mathbf{A}_4 \end{bmatrix}, \quad \tilde{\mathbf{B}} = \mathbf{T} \mathbf{B} = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{O} \end{bmatrix},$$

where $\mathbf{A}_3 = \mathbf{O}$. It is clear that $\mathbf{T} \mathbf{B}$ must have the above form. If $\mathbf{A}_3$ were not a zero matrix, then we would have

$$\tilde{\mathbf{A}} \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{A}_1 \mathbf{B}_1 \\ \mathbf{A}_3 \mathbf{B}_1 \end{bmatrix}.$$

Comparing the above with (3.15), we conclude that $\mathbf{A}_3 \mathbf{B}_1 = \mathbf{O}$. Thus,

$$\tilde{\mathbf{A}} \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{A}_1 \mathbf{B}_1 \\ \mathbf{O} \end{bmatrix}.$$

Next we consider

$$\tilde{\mathbf{A}}^2 \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{A}_1^2 \mathbf{B}_1 \\ \mathbf{A}_3 \mathbf{A}_1 \mathbf{B}_1 \end{bmatrix}.$$

Again, comparing the above with (3.15), we conclude that

$$\mathbf{A}_3 \mathbf{A}_1 \mathbf{B}_1 = \mathbf{O},$$

and thus

$$\tilde{\mathbf{A}}^2 \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{A}_1^2 \mathbf{B}_1 \\ \mathbf{O} \end{bmatrix}.$$

Continuing in this manner, we conclude that $\mathbf{A}_3 \mathbf{A}_1^{n-1} \mathbf{B}_1 = \mathbf{O}$. Thus it follows that

$$\mathbf{A}_3 [\mathbf{B}_1 \quad \mathbf{A}_1 \mathbf{B}_1 \quad \cdots \quad \mathbf{A}_1^{n-1} \mathbf{B}_1] = \mathbf{O}.$$

Because $\text{rank}[\mathbf{B}_1 \quad \mathbf{A}_1 \mathbf{B}_1 \quad \cdots \quad \mathbf{A}_1^{n-1} \mathbf{B}_1] = r$ —that is, the controllability matrix of the pair $(\mathbf{A}_1, \mathbf{B}_1)$ is of full rank—we have to have $\mathbf{A}_3 = \mathbf{O}$, and the proof of necessity is complete.

To prove sufficiency ($\Leftarrow$), note that if there is a similarity transformation such that

$$\tilde{\mathbf{A}} = \mathbf{T} \mathbf{A} \mathbf{T}^{-1} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{O} & \mathbf{A}_4 \end{bmatrix}, \quad \tilde{\mathbf{B}} = \mathbf{T} \mathbf{B} = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{O} \end{bmatrix},$$

then the controllability matrix of the pair $(\tilde{\mathbf{A}}, \tilde{\mathbf{B}})$ has the form

$$\begin{bmatrix} \mathbf{B}_1 & \mathbf{A}_1 \mathbf{B}_1 & \mathbf{A}_1^2 \mathbf{B}_1 & \cdots & \mathbf{A}_1^{n-1} \mathbf{B}_1 \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \cdots & \mathbf{O} \end{bmatrix}.$$

Hence, the pair $(\tilde{\mathbf{A}}, \tilde{\mathbf{B}})$ is nonreachable. Because the similarity transformation preserves the reachability property, the pair $(\mathbf{A}, \mathbf{B})$ is nonreachable as well.

◆ Example 3.3

For the system model,

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{u},$$

construct a state variable transformation so that in the new coordinates the reachable part is separated and distinct from the nonreachable part.

We first form the controllability matrix of the pair $(\mathbf{A}, \mathbf{B})$ to get

$$\mathbf{C} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \mathbf{A}^3\mathbf{B}] = \begin{bmatrix} 1 & 1 & -1 & 0 & 3 & 4 & -1 & 4 \\ 0 & 0 & 1 & 1 & 0 & 1 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 7 & 7 & 19 \end{bmatrix}.$$

We then use the MATLAB command $[\mathbf{Q} \ \mathbf{R}] = \text{qr}(\mathbf{C})$ to obtain

$$\mathbf{Q} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} -1 & -1 & 1 & 0 & -3 & -4 & 1 & -4 \\ 0 & -1 & -1 & -3 & -2 & -7 & -7 & -19 \\ 0 & 0 & -1 & -1 & 0 & -1 & -3 & -5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where the matrices $\mathbf{Q}$ and $\mathbf{R}$ satisfy $\mathbf{C} = \mathbf{QR}$, and $\mathbf{R}$ is upper row triangular such that $\text{rank } \mathbf{R} = \text{rank } \mathbf{C}$. Thus, premultiplying the controllability matrix, $\mathbf{C}$, by $\mathbf{Q}^{-1}$ reduces this matrix to an upper row triangular matrix $\mathbf{R}$ because $\mathbf{Q}^{-1}\mathbf{C} = \mathbf{R}$. Let $\mathbf{z} = \mathbf{Q}^{-1}\mathbf{x}$ be the state variable transformation. Then in the new coordinates the matrices $\mathbf{A}$ and $\mathbf{B}$ take the form

$$\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \left[\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 4 \end{array} \right] \quad \text{and} \quad \mathbf{Q}^{-1}\mathbf{B} = \left[\begin{array}{c|c} -1 & -1 \\ 0 & -1 \\ \hline 0 & 0 \\ 0 & 0 \end{array} \right].$$

The dimension of the nonreachable part is 1.

If the eigenvalues of the noncontrollable part of a given pair $(\mathbf{A}, \mathbf{B})$ are all in the open left-hand complex plane—that is, the nonreachable part is asymptotically stable—then the pair $(\mathbf{A}, \mathbf{B})$ is called *stabilizable*.

We now give yet another test for reachability. This test is also known as the Popov–Belevitch–Hautus (PBH) eigenvector test. We denote the set of eigenvalues of a matrix $\mathbf{A}$ as $\text{eig}(\mathbf{A})$.

Theorem 3.5 The pair $(\mathbf{A}, \mathbf{B})$ is reachable if and only if

$$\text{rank}[s\mathbf{I}_n - \mathbf{A} \quad \mathbf{B}] = n \quad \text{for all } s \in \text{eig}(\mathbf{A}).$$

Proof We first prove the necessity part ($\Rightarrow$) by contraposition. If there is a vector $\mathbf{q} \neq \mathbf{0}$ such that for some $s \in \text{eig}(\mathbf{A})$,

$$\mathbf{q}^T [s\mathbf{I}_n - \mathbf{A} \quad \mathbf{B}] = \mathbf{0}^T,$$

that is,

$$\mathbf{q}^T \mathbf{A} = s\mathbf{q}^T, \quad \mathbf{q}^T \mathbf{B} = \mathbf{0}^T,$$

then

$$\begin{aligned} \mathbf{q}^T \mathbf{A} \mathbf{B} &= s\mathbf{q}^T \mathbf{B} = \mathbf{0}^T, \\ \mathbf{q}^T \mathbf{A}^2 \mathbf{B} &= s\mathbf{q}^T \mathbf{A} \mathbf{B} = \mathbf{0}^T, \end{aligned}$$

and so on, until

$$\mathbf{q}^T \mathbf{A}^{n-1} \mathbf{B} = s\mathbf{q}^T \mathbf{A}^{n-2} \mathbf{B} = \mathbf{0}^T.$$

We write the above equations in the form

$$\mathbf{q}^T [\mathbf{B} \quad \mathbf{A} \mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1} \mathbf{B}] = \mathbf{0}^T,$$

which means that the pair $(\mathbf{A}, \mathbf{B})$ is nonreachable.

We now prove the sufficiency part ($\Leftarrow$) also by contraposition. So we assume that $\text{rank}[\mathbf{B} \quad \mathbf{A} \mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1} \mathbf{B}] = r < n$. By Theorem 3.4, there exists a state variable transformation such that in the new basis the pair $(\mathbf{A}, \mathbf{B})$ takes the form

$$\tilde{\mathbf{A}} = \mathbf{T} \mathbf{A} \mathbf{T}^{-1} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{O} & \mathbf{A}_4 \end{bmatrix}, \quad \tilde{\mathbf{B}} = \mathbf{T} \mathbf{B} = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{O} \end{bmatrix}.$$

We choose $\mathbf{q}$ of the form

$$\mathbf{q}^T = [\mathbf{0} \quad \mathbf{z}],$$

where $\mathbf{z} \neq \mathbf{0}$ is a left eigenvector of $\mathbf{A}_4$; that is,

$$\mathbf{z} \mathbf{A}_4 = s\mathbf{z},$$

where $s \in \text{eig}(\mathbf{A}_4)$. Thus

$$\begin{aligned} \mathbf{q}^T \tilde{\mathbf{A}} &= [\mathbf{0} \quad \mathbf{z}] \tilde{\mathbf{A}} \\ &= [\mathbf{0} \quad s\mathbf{z}] \\ &= s\mathbf{q}^T. \end{aligned}$$

Clearly,

$$\mathbf{q}^T [s\mathbf{I}_n - \tilde{\mathbf{A}} \quad \tilde{\mathbf{B}}] = \mathbf{0}^T,$$

which means that the rank of the matrix $[s\mathbf{I}_n - \tilde{\mathbf{A}} \quad \tilde{\mathbf{B}}]$ is less than n , and therefore the proof of the sufficiency part is complete.

3.2 Observability and Constructability

As we have seen in the previous section, system reachability implies the ability to completely control the entire state of the system. Suppose now that the system state is not directly accessible. Instead, we have the output of the system,

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u},$$

where $\mathbf{C} \in \mathbb{R}^{p \times n}$, $p \leq n$, and $\mathbf{D} \in \mathbb{R}^{p \times m}$. However, we still want to know the dynamical behavior of the entire state.

We say that the system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u},$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u},$$

or equivalently the pair $(\mathbf{A}, \mathbf{C})$, is **observable** if there is a finite $t_1 > t_0$ such that **for arbitrary $\mathbf{u}(t)$ and resulting $\mathbf{y}(t)$ over $[t_0, t_1]$, we can determine $\mathbf{x}(t_0)$** from complete knowledge of the system input $\mathbf{u}$ and output $\mathbf{y}$.

Note that once $\mathbf{x}(t_0)$ is known, we can determine $\mathbf{x}(t)$ from knowledge of $\mathbf{u}(t)$ and $\mathbf{y}(t)$ over any finite time interval $[t_0, t_1]$. We now show how to constructively determine $\mathbf{x}(t_0)$, given $\mathbf{u}(t)$ and $\mathbf{y}(t)$. While doing so we will establish a sufficiency condition for the system to be observable. The condition is also necessary for the system observability, as we show next. To proceed with constructive determination of the state vector, we note that the solution $\mathbf{y}(t)$ is given by

$$\mathbf{y}(t) = \mathbf{C}e^{\mathbf{A}(t-t_0)}\mathbf{x}_0 + \int_{t_0}^t \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau + \mathbf{D}\mathbf{u}(t).$$

We subtract $\int_{t_0}^t \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau + \mathbf{D}\mathbf{u}(t)$ from both sides of the above equation. Let

$$\mathbf{g}(t) = \mathbf{y}(t) - \int_{t_0}^t \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau - \mathbf{D}\mathbf{u}(t),$$

then we have

$$\mathbf{g}(t) = \mathbf{C}e^{\mathbf{A}(t-t_0)}\mathbf{x}(t_0),$$

where $\mathbf{g}$ is known to us. Our goal now is to determine $\mathbf{x}(t_0)$. Having $\mathbf{x}(t_0)$, we can determine the entire $\mathbf{x}(t)$ for all $t \in [t_0, t_1]$ from the formula

$$\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)}\mathbf{x}_0 + \int_{t_0}^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau.$$

Premultiplying both side of $\mathbf{g}(t)$ by $e^{\mathbf{A}^T(t-t_0)}\mathbf{C}^T$ and integrating the resulting expression between the limits t_0 and t_1 , we get

$$\int_{t_0}^{t_1} e^{\mathbf{A}^T(t-t_0)}\mathbf{C}^T\mathbf{C}e^{\mathbf{A}(t-t_0)} dt \mathbf{x}_0 = \int_{t_0}^{t_1} e^{\mathbf{A}^T(t-t_0)}\mathbf{C}^T \mathbf{g}(t) dt.$$

Let

$$V(t_0, t_1) = \int_{t_0}^{t_1} e^{A^T t} \mathbf{C}^T \mathbf{C} e^{A t} dt.$$

Then, after performing some manipulations and assuming that the matrix $V(t_0, t_1)$ is invertible, we obtain

$$\mathbf{x}_0 = e^{A t_0} V^{-1}(t_0, t_1) e^{A^T t_0} \int_{t_0}^{t_1} e^{A^T(t-t_0)} \mathbf{C}^T \mathbf{g}(t) dt.$$

Knowledge of $\mathbf{x}_0$ allows us to reconstruct the entire state $\mathbf{x}(t)$ over the interval $[t_0, t_1]$. We conclude that if the matrix $V(t_0, t_1)$ is invertible, then the system is observable.

◆ Example 3.4

For a dynamical system modeled by

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} -1/2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u = \mathbf{A}\mathbf{x} + \mathbf{b}u, \\ y &= [1 \quad 1] \mathbf{x} = \mathbf{c}\mathbf{x}, \end{aligned}$$

we shall determine $\mathbf{x}(t)$ over the time interval $[0, 10]$ knowing that $u(t) = 1$ for $t \geq 0$ and $y(t) = t^2 + 2t$ for $t \geq 0$.

We first find $\mathbf{x}(0)$. Knowledge of $\mathbf{x}(0)$ will allow us to find the entire state vector for all $t \in [0, 10]$. We have $t_0 = 0$ and $t_1 = 10$. Hence,

$$\begin{aligned} \mathbf{x}(t_0) = \mathbf{x}(0) &= e^{A t_0} V^{-1}(t_0, t_1) e^{A^T t_0} \int_{t_0}^{t_1} e^{A^T(t-t_0)} \mathbf{c}^T g(t) dt \\ &= V^{-1}(0, 10) \int_0^{10} e^{A^T t} \mathbf{c}^T g(t) dt, \end{aligned}$$

where

$$g(t) = y(t) - \mathbf{c} \int_0^t e^{A(t-\tau)} \mathbf{b}u(\tau) d\tau.$$

To evaluate the above expression, we first use formula (A.148) to compute

$$\begin{aligned} e^{A t} &= \mathcal{L}^{-1}([s \mathbf{I}_2 - \mathbf{A}]^{-1}) = \mathcal{L}^{-1} \left(\begin{bmatrix} \frac{1}{s} - \frac{1}{2} \frac{1}{s^2} & \frac{1}{2} \frac{1}{s^2} \\ -\frac{1}{2} \frac{1}{s^2} & \frac{1}{s} + \frac{1}{2} \frac{1}{s^2} \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 - \frac{1}{2}t & \frac{1}{2}t \\ -\frac{1}{2}t & 1 + \frac{1}{2}t \end{bmatrix}. \end{aligned}$$

Next, we compute

$$\begin{aligned}
 \mathbf{V}(0, 10) &= \int_0^{10} e^{\mathbf{A}^T t} \mathbf{c}^T \mathbf{c} e^{\mathbf{A} t} dt \\
 &= \int_0^{10} \left(\begin{bmatrix} 1-t \\ 1+t \end{bmatrix} [1-t \quad 1+t] \right) dt = \int_0^{10} \begin{bmatrix} 1-2t+t^2 & 1-t^2 \\ 1-t^2 & 1+2t+t^2 \end{bmatrix} dt \\
 &= \left[\begin{array}{cc} t-t^2+\frac{t^3}{3} & t-\frac{t^3}{3} \\ t-\frac{t^3}{3} & t+t^2+\frac{t^3}{3} \end{array} \right] \bigg|_0^{10} = \begin{bmatrix} 243.333 & -323.333 \\ -323.333 & 443.333 \end{bmatrix}.
 \end{aligned}$$

The inverse of $\mathbf{V}(0, 10)$ is

$$\mathbf{V}^{-1}(0, 10) = \begin{bmatrix} 0.133 & 0.097 \\ 0.097 & 0.073 \end{bmatrix}.$$

We can now compute $g(t)$ to get

$$\begin{aligned}
 g(t) &= y(t) - \mathbf{c} \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{b} u(\tau) d\tau \\
 &= t^2 + 2t - [1 \quad 1] \int_0^t \begin{bmatrix} -1+t-\tau \\ 1+t+\tau \end{bmatrix} d\tau = t^2 + 2t - \int_0^t (2t-2\tau) d\tau \\
 &= t^2 + 2t - 2t(\tau) \bigg|_0^t + \tau^2 \bigg|_0^t = t^2 + 2t - t^2 \\
 &= 2t.
 \end{aligned}$$

Hence,

$$\int_0^{10} e^{\mathbf{A}^T t} \mathbf{c}^T g(t) dt = \int_0^{10} \begin{bmatrix} (1-t)2t \\ (1+t)2t \end{bmatrix} dt = \left[\begin{array}{c} t^2 - \frac{2}{3}t^3 \\ t^2 + \frac{2}{3}t^3 \end{array} \right] \bigg|_0^{10} = \begin{bmatrix} -566.6667 \\ 766.6667 \end{bmatrix}.$$

We can now compute

$$\mathbf{x}(0) = \mathbf{V}^{-1}(0, 10) \int_0^{10} e^{\mathbf{A}^T t} \mathbf{c}^T g(t) dt = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Alternatively, we could use the Laplace transform method to find $\mathbf{x}(0)$. Knowing $\mathbf{x}(0)$, we can find $\mathbf{x}(t)$ for $t \geq 0$,

$$\begin{aligned}
 \mathbf{x}(t) &= e^{\mathbf{A}t} \mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{b} u(\tau) d\tau \\
 &= \begin{bmatrix} -1+t \\ 1+t \end{bmatrix} + \begin{bmatrix} (-1+t)t - \frac{t^2}{2} \\ (1+t)t - \frac{t^2}{2} \end{bmatrix} = \begin{bmatrix} \frac{t^2}{2} - 1 \\ \frac{t^2}{2} + 2t + 1 \end{bmatrix}.
 \end{aligned}$$

We will now show that the invertibility of $V(t_0, t_1)$ is also necessary for observability. We first note that one can use similar arguments to those in the previous section to show that $V(t_0, t_1)$ is nonsingular if and only if the n columns of the matrix

$$Ce^{At}$$

are linearly independent for all $t \in [0, \infty)$ over the set of real numbers. Hence, if $V(t_0, t_1)$ is singular, there exists a nonzero constant vector, say $\mathbf{x}_a$, such that $V(t_0, t_1)\mathbf{x}_a = \mathbf{0}$. This implies that

$$\mathbf{g}(t) = Ce^{At}\mathbf{x}_0 = Ce^{At}(\mathbf{x}_0 + \mathbf{x}_a).$$

Thus, $\mathbf{x}(0) = \mathbf{x}_0 + \mathbf{x}_a$ yields the same response as $\mathbf{x}(0) = \mathbf{x}_0$, which means that we cannot determine the system state, or in other words, the state fails to be observable. In view of the above result, it follows that by repeating essentially the same arguments used in the previous section we can prove the following theorem:

Theorem 3.6 The following statements are equivalent:

1. The pair $(\mathbf{A}, \mathbf{C})$ is observable.
2. The matrix $V(t_0, t_1)$ is nonsingular for all $t_1 > t_0$.
3. The n columns of Ce^{At} are linearly independent for all $t \in [0, \infty)$ over the real numbers.
4. The observability matrix

$$\begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \in \mathbb{R}^{pn \times n}$$

5. is of full rank n .

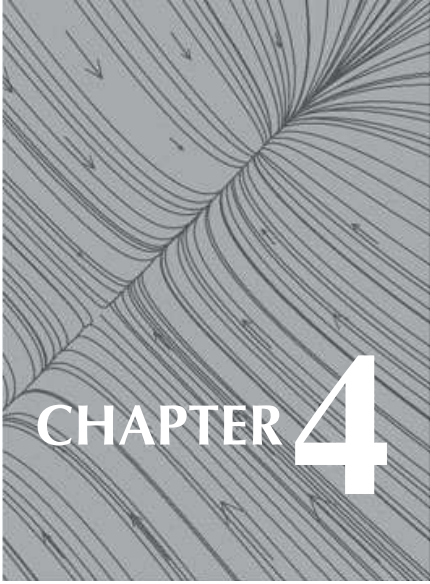
$$\text{rank} \begin{bmatrix} s\mathbf{I}_n - \mathbf{A} \\ \mathbf{C} \end{bmatrix} = n \quad \text{for all } s \in \text{eig}(\mathbf{A}).$$

◆ Example 3.5

For the dynamical system model,

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{Ax} = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix} \mathbf{x}, \\ y &= \mathbf{cx} = [-1 \quad 1] \mathbf{x}, \end{aligned}$$

we will find a nonzero initial vector $\mathbf{x}(0) = \mathbf{x}_0$ such that $y(t) = 0$ for all $t \geq 0$.



CHAPTER 4

Stability

Examine each question in terms of what is ethically and aesthetically right, as well as what is economically expedient. A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong when it tends otherwise.

—Aldo Leopold

4.1 Informal Introduction to Stability

In general, it can be difficult to obtain an explicit solution of nonlinear differential equations modeling a dynamical system. However, often we do not need to know the explicit solution of the modeling equations. Instead, we are interested in the behavior of the solution as time tends to infinity. Furthermore, in many models of physical systems a “small” change in the initial conditions results in a “small” change in the solution. Specifically, if a system is perturbed from its rest, or equilibrium position, then it starts to move. We can roughly say that the equilibrium position is stable if the system does not go far from this position for small initial perturbations. This concept may be illustrated, as in Willems [300], Brogan [35], or Barnett [20], by means of a ball resting in equilibrium on a sheet of metal bent into various shapes with cross sections depicted in Figure 4.1. This figure illustrates four possible scenarios. If we neglect frictional forces, a small perturbation from the equilibrium leads to the following:

1. **Oscillatory** motion about equilibrium **irrespective of the magnitude** of the initial perturbation. In this case the equilibrium position is globally stable.
2. The **new position** is also an **equilibrium**, and the ball will remain in it. Here the equilibrium position is stable.
3. The **ball moving away without returning to the equilibrium**. This equilibrium position is unstable.
4. **Oscillatory** motion about the equilibrium, **unless the initial perturbation is so large** that the ball is forced to oscillate about the new equilibrium position. This equilibrium position is locally stable.

If friction is taken into account, then the oscillatory motion steadily decreases until the ball returns to the equilibrium positions as in items 1 and 4 above. We refer to such equilibrium positions as being asymptotically stable. In our discussion of stability concepts, we implicitly

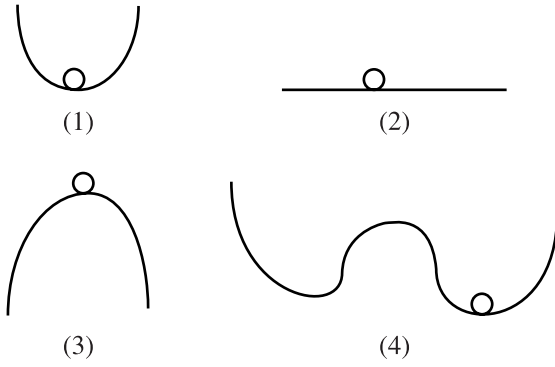


Figure 4.1 Illustration of stable and unstable equilibrium positions.

assume, as in item 4, that there is some “small” region for allowed initial perturbations for which subsequent motions converge to the equilibrium position. If however, as in item 1, any initial perturbation yields a motion that converges to the equilibrium position, then we have global asymptotic stability, which is also referred to as asymptotic stability in the large.

The above informal discussion should serve as a preparation for an in-depth analysis of stability of dynamical systems. We begin our analysis by formally defining the equilibrium state. We then define stability, asymptotic stability, and other relevant notions.

4.2 Basic Definitions of Stability

We consider a class of dynamical systems modeled by the equation:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t)), \quad \mathbf{x}(t_0) = \mathbf{x}_0, \quad (4.1)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ is the state vector and $\mathbf{f} : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a vector-valued function with the components

$$f_i(t, x_1, x_2, \dots, x_n) : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}, \quad i = 1, 2, \dots, n.$$

We assume that f_i 's are continuous and have continuous first partial derivatives. Under these assumptions, the solution to $\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t))$, $\mathbf{x}(t_0) = \mathbf{x}_0$, exists and is unique. We denote this solution as $\mathbf{x}(t) = \mathbf{x}(t; t_0, \mathbf{x}_0)$. If the f_i 's do not depend explicitly on t , then the system represented by $\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t)) = \mathbf{f}(\mathbf{x}(t))$ is called *autonomous*; otherwise it is called *nonautonomous*. An *equilibrium point* or state is a constant vector, say $\mathbf{x}_e$, such that

$$\mathbf{f}(t, \mathbf{x}_e) = \mathbf{0} \quad \text{for all } t.$$

Observe that an equilibrium state $\mathbf{x}_e$ is a constant solution to (4.1).

If $\mathbf{x}_e$ is an equilibrium point, we can arrange for this equilibrium point to be transferred to the origin of $\mathbb{R}^n$ by introducing the new variable:

$$\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_e.$$

We assume that this has been done for an equilibrium point under consideration. Therefore, we have

$$\mathbf{f}(t, \mathbf{0}) = \mathbf{0} \quad \text{for all } t.$$

We now introduce basic definitions of stability in the sense of A. M. Lyapunov* (1857–1918). In the following considerations we use the standard Euclidean norm of a vector. Thus, if $\mathbf{x} \in \mathbb{R}^n$, then

$$\|\mathbf{x}\| = \|\mathbf{x}\|_2 = (\mathbf{x}^T \mathbf{x})^{1/2}.$$

Definition 4.1 An equilibrium state $\mathbf{x}_e$ is said to be **stable** if for any given t_0 and any positive scalar ε , there exists a positive scalar $\delta = \delta(t_0, \varepsilon)$ such that if $\|\mathbf{x}(t_0) - \mathbf{x}_e\| < \delta$, then $\|\mathbf{x}(t; t_0, \mathbf{x}_0) - \mathbf{x}_e\| < \varepsilon$ for all $t \geq t_0$.

Friedland [91, p. 43] makes the following comment regarding the above definition of stability: “If this ‘epsilonics’ definition leaves you cold, think of a contest between you, the system designer, and an adversary (nature?). The adversary picks a region in state space of radius ε and challenges you to find another region, of radius δ , such that if the initial state starts out inside your region, it remains inside his region. If you can do this, your system design is stable.” We add that you have to find an appropriate δ for any choice of ε for the equilibrium state to be stable.

Definition 4.2 An equilibrium state $\mathbf{x}_e$ is said to be **convergent**, or *attractive*, if for any given t_0 there is a positive scalar $\delta_1 = \delta_1(t_0)$ such that if $\|\mathbf{x}(t_0) - \mathbf{x}_e\| < \delta_1$, then

$$\lim_{t \rightarrow \infty} \mathbf{x}(t; t_0, \mathbf{x}_0) = \mathbf{x}_e.$$

We can restate Definition 4.2 as follows. An equilibrium state $\mathbf{x}_e$ is convergent if for any t_0 there is a positive scalar $\delta_1 = \delta_1(t_0)$ such that if $\|\mathbf{x}(t_0) - \mathbf{x}_e\| < \delta_1$, then for any given $\varepsilon_1 > 0$ there exists a **$T = T(\varepsilon_1, t_0, \mathbf{x}_0)$** such that **$\|\mathbf{x}(t; t_0, \mathbf{x}_0) - \mathbf{x}_e\| < \varepsilon_1$** for all $t > t_0 + T$.

Definition 4.3 An equilibrium state $\mathbf{x}_e$ is **asymptotically stable** if it is **stable and convergent**.

Definition 4.4 An equilibrium state $\mathbf{x}_e$ is said to be **uniformly stable** if the positive scalar **$\delta = \delta(t_0, \varepsilon)$** introduced in Definition 4.1 can be taken **independent of the initial time t_0** .

◆ Example 4.1

The following example that comes from Rugh [245, p. 100] nicely illustrates the notion of uniform stability. The solution to the differential equation,

$$\dot{x}(t) = (4t \sin t - 2t)x(t), \quad x(t_0) = x_0, \quad (4.2)$$

is

$$x(t) = x_0 \exp(4 \sin t - 4t \cos t - t^2 - 4 \sin t_0 + 4t_0 \cos t_0 + t_0^2). \quad (4.3)$$

One can verify this using MATLAB's Symbolic Toolbox by typing the command

```
x=dsolve('Dx=(4*t*sin(t)-2*t)*x', 'x(t_0)=x_0', 't')
```

It is easy to show that for fixed t_0 there is a γ such that (4.3) is bounded by $\gamma|x_0|$ for all $t \geq t_0$. This is because the term $(-t^2)$ dominates the exponent as t increases. Therefore, for a given ε , it is enough to choose x_0 so that $\gamma|x_0| < \varepsilon$, or equivalently $\delta < \varepsilon/\gamma$, to show that the equilibrium solution $x_e = 0$ is stable. However, the

*Sometimes transliterated in the literature as Liapunov or Liapunoff.

equilibrium solution $x_e = 0$ is not uniformly stable. To show this, fix initial state x_0 and consider a sequence of initial times $t_0 = 2k\pi$, where $k = 0, 1, \dots$, and the values of the respective solutions at times π units later:

$$x(2k\pi + \pi) = x_0 \exp((4k + 1)\pi(4 - \pi)).$$

One can see that there is no bound on the exponential factor and hence a candidate bound γ increases as k increases, where increasing k corresponds to the increasing initial time.

Definition 4.5 An equilibrium state x_e is said to be **uniformly convergent**, or uniformly attractive, if the positive scalars $\delta_1 = \delta_1(t_0)$ and $T = T(\varepsilon_1, t_0, x_0)$ introduced in Definition 4.2 are **independent of t_0** .

The last two definitions allow us to introduce the notion of uniform asymptotic stability.

Definition 4.6 An equilibrium state x_e is said to be **uniformly asymptotically stable** if it is **uniformly stable and uniformly convergent**.

Notice that the solution $x(t) = x_e$ must first be stable in order to qualify to be asymptotically stable. This is to prevent a system trajectory from straying arbitrarily far from the equilibrium state before converging toward it. Note also that the concepts of Lyapunov stability and asymptotic stability are local concepts. They apply to the system's behavior in a small neighborhood about the equilibrium point (Hsu and Meyer [128]).

A special case of uniform asymptotic stability is **uniform exponential stability that implies uniform stability**, and it imposes an additional requirement that all solutions originating “close” to the given equilibrium, $x_e = 0$, approach this equilibrium exponentially as $t \rightarrow \infty$. More rigorously, we say that $x_e = 0$ is uniformly exponentially stable if it is stable and there exist finite positive constants γ and λ such that for any t_0 and x_0 , any solution starting in a neighborhood of 0 satisfies

$$\|x(t)\| \leq \gamma e^{-\lambda(t-t_0)} \|x_0\|.$$

Note that γ is no less than unity, and uniform implies that γ and λ are independent of t_0 (see Figure 4.2).

An equilibrium state is unstable if it is not stable. An ε - δ definition of instability is given next.

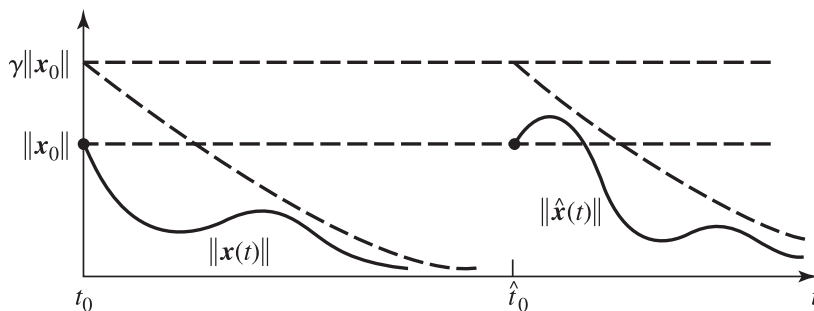


Figure 4.2 Illustration of uniform exponential stability: A decaying-exponential bound is independent of the initial time t_0 .

Definition 4.7 An equilibrium state $\mathbf{x}_e$ is said to be unstable if there is an $\varepsilon > 0$ so that for any $\delta > 0$ there exists $\mathbf{x}(t_0)$ such that if $\|\mathbf{x}(t_0) - \mathbf{x}_e\| < \delta$, then $\|\mathbf{x}(t_1) - \mathbf{x}_e\| \geq \varepsilon$ for some $t_1 > t_0$.

A two-dimensional geometric interpretation of the above definitions is shown in Figure 4.3. We assume that $\mathbf{x}_e = 0$ is the equilibrium state. If the equilibrium state is stable, as in Figure 4.3(a), then given an outer circle of radius ε , there exists an inner circle of radius δ such that trajectories starting within the δ -circle never leave the ε -circle. If the equilibrium state is asymptotically stable, then trajectories tend to the equilibrium state as t tends to ∞ . This situation is depicted in Figure 4.3(b). Finally, if the equilibrium state is unstable, as in Figure 4.3(c), then there exists an ε -circle such that for every δ -circle there exists a trajectory starting within it that leaves the ε -circle at some later time. A geometric interpretation of the above described scenarios in one dimension is given in Figure 4.4.

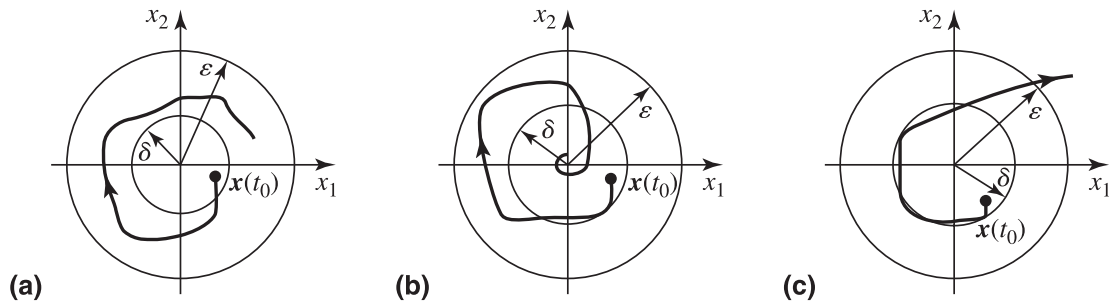


Figure 4.3 Illustration of stability concepts in two dimensions. (a) **Stable** equilibrium state in the sense of Lyapunov. (b) **Asymptotically stable** equilibrium state. (c) **Unstable** equilibrium state.

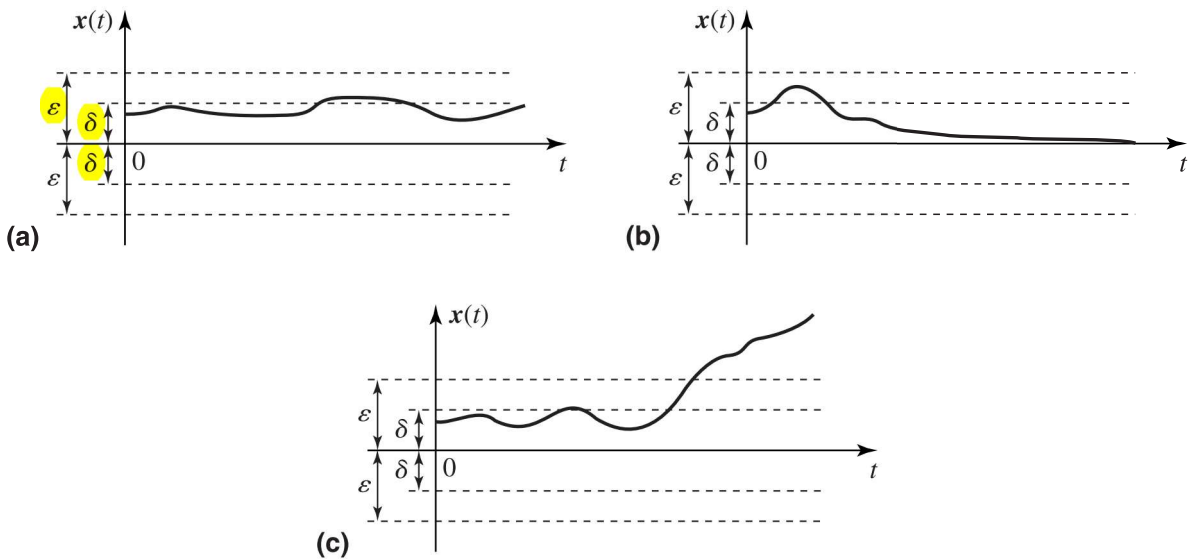


Figure 4.4 Illustration of stability concepts in one dimension. (a) **Stable** equilibrium state in the sense of Lyapunov. (b) **Asymptotically stable** equilibrium state. (c) **Unstable** equilibrium state.

4.3 Stability of Linear Systems

It follows from the discussion in the previous sections that a linear time-invariant dynamical system modeled by the equations

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{x}_0 \quad (4.4)$$

is asymptotically stable if and only if the solution to equation (4.4) decays to zero as $t \rightarrow \infty$ for any initial state $\mathbf{x}_0$. We can view the vector $\mathbf{x}(t) \in \mathbb{R}^n$ as defining the coordinates of a point moving in an n -dimensional state space $\mathbb{R}^n$. In an asymptotically stable system modeled by equation (4.4), this point converges to the origin of $\mathbb{R}^n$. We can argue that if a trajectory is converging to the origin of the state space, then it should be possible to find a family of nested surfaces described by $V(x_1, x_2, \dots, x_n) = c$, $c \geq 0$, such that monotonically decreasing values of c give surfaces progressively shrinking in on the origin with the limiting surface $V(x_1, x_2, \dots, x_n) = 0$ being the origin $\mathbf{x} = \mathbf{0}$. Furthermore, along every trajectory of (4.4), the parameter c should steadily decrease. We now introduce some notation and definitions that we use in the ensuing discussion.

Let $V(\mathbf{x}) = V(x_1, x_2, \dots, x_n) : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function and let $\mathcal{S} \subseteq \mathbb{R}^n$ be a compact region containing the origin $\mathbf{x} = \mathbf{0}$ in its interior.

Definition 4.8 We say that the function $V = V(\mathbf{x})$ is *positive semidefinite* (p.s.d.) in $\mathcal{S}$, with respect to $\mathbf{x} = \mathbf{0}$, if:

1. V is continuously differentiable, that is, $V \in \mathcal{C}^1$.
2. $V(\mathbf{0}) = 0$.
3. $V(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathcal{S}$.

Definition 4.9 We say that the function $V = V(\mathbf{x})$ is *positive definite* (p.d.) in $\mathcal{S}$, with respect to $\mathbf{x} = \mathbf{0}$, if:

1. $V \in \mathcal{C}^1$.
2. $V(\mathbf{0}) = 0$.
3. $V(\mathbf{x}) > 0$ for all $\mathbf{x} \in \mathcal{S} \setminus \{\mathbf{0}\}$.

The symbol $\mathcal{S} \setminus \{\mathbf{0}\}$ in item 3 above denotes the set $\{\mathbf{x} : \mathbf{x} \in \mathcal{S}, \mathbf{x} \neq \mathbf{0}\}$.

Negative semidefinite (n.s.d.) and *negative definite* (n.d.) functions are defined in a similar fashion by reversing inequality signs in item 3 in both definitions above.

There are many functions that satisfy the above definitions. In the stability analysis of dynamical systems, one class is particularly useful. This is the positive definite quadratic form, a label attached to a nested family of ellipses and their extensions to higher-dimensional spaces—ellipsoids for $n = 3$ and hyperellipsoids for $n \geq 4$. Quadratic forms are reviewed in Section A.4. We consider the quadratic form

$$V(\mathbf{x}) = \mathbf{x}^T \mathbf{P} \mathbf{x},$$

where $\mathbf{P}$ is a real symmetric $n \times n$ matrix. The time derivative of $V(\mathbf{x}(t))$ evaluated on a solution of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is

$$\begin{aligned} \frac{dV(\mathbf{x}(t))}{dt} &= \dot{V}(t) \\ &= \dot{\mathbf{x}}^T(t) \mathbf{P} \mathbf{x}(t) + \mathbf{x}^T(t) \mathbf{P} \dot{\mathbf{x}}(t). \end{aligned}$$

Substituting $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ and $\dot{\mathbf{x}}^T = \mathbf{x}^T \mathbf{A}^T$ into the above equation yields

$$\begin{aligned}\dot{V} &= \mathbf{x}^T \mathbf{A}^T \mathbf{P} \mathbf{x} + \mathbf{x}^T \mathbf{P} \mathbf{A} \mathbf{x} \\ &= \mathbf{x}^T (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A}) \mathbf{x}.\end{aligned}$$

Let

$$\mathbf{Q} = -(\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A}).$$

Then,

$$\dot{V} = -\mathbf{x}^T \mathbf{Q} \mathbf{x}. \quad (4.5)$$

We are now ready to state and prove the following theorem concerning the stability of the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, or, equivalently, the matrix $\mathbf{A} \in \mathbb{R}^n$.

Theorem 4.1 (A. M. Lyapunov, 1892) The system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, $\mathbf{x}(t_0) = \mathbf{x}_0$, is asymptotically stable if and only if for any given real symmetric positive definite (r.s.p.d.) matrix $\mathbf{Q}$ the solution $\mathbf{P}$ to the continuous Lyapunov matrix equation,

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q} \quad (4.6)$$

is also (real symmetric) positive definite.

Proof ($\Rightarrow$) We prove necessity, for asymptotic stability, by contradiction. We have $-\mathbf{x}^T \mathbf{Q} \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$. We assume that $\mathbf{A}$ is an asymptotically stable matrix and that for some nonzero $\mathbf{x}_0$ we have $\mathbf{x}_0^T \mathbf{P} \mathbf{x}_0 \leq 0$. From the discussion before the theorem, and in particular from (4.5), it follows that

$$\frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) = -\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) < 0.$$

The above means that $V(\mathbf{x}(t)) = \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)$ decreases on the trajectories of the system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ as $t \rightarrow \infty$. On the other hand, because $\mathbf{A}$ is asymptotically stable, we have

$$\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{0}.$$

Hence,

$$\lim_{t \rightarrow \infty} \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) = \lim_{\mathbf{x}(t) \rightarrow \mathbf{0}} \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) = 0.$$

Thus, we arrived at a contradiction. This completes the proof of the first part of the theorem that states that asymptotic stability of $\mathbf{A}$ implies that for any r.s.p.d matrix $\mathbf{Q}$ the solution $\mathbf{P}$ to the equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q}$ is r.s.p.d..

($\Leftarrow$) We also prove sufficiency by contradiction. We assume that

$$\mathbf{x}^T \mathbf{P} \mathbf{x} > 0 \quad \text{and} \quad -\mathbf{x}^T \mathbf{Q} \mathbf{x} < 0 \quad \text{for all } \mathbf{x} \neq \mathbf{0},$$

and that the matrix $\mathbf{A}$ is not asymptotically stable. This means that the solution to $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$, $\mathbf{x}_0 \neq \mathbf{0}$, which has the form $\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)} \mathbf{x}_0$, is bounded away from the origin $\mathbf{x} = \mathbf{0}$. Thus, there is a constant $b > 0$ such that

$$\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) > b > 0 \quad \text{for all } t > t_0.$$

Integrating $\frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) = -\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t)$ yields

$$\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) = \mathbf{x}^T(t_0) \mathbf{P} \mathbf{x}(t_0) + \int_{t_0}^t (-\mathbf{x}^T(s) \mathbf{Q} \mathbf{x}(s)) ds.$$

Taking into account that $-\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) < -b < 0$ for all $t > t_0$, we obtain for $t > t_0$,

$$\begin{aligned} \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) &= \mathbf{x}^T(t_0) \mathbf{P} \mathbf{x}(t_0) + \int_{t_0}^t (-\mathbf{x}^T(s) \mathbf{Q} \mathbf{x}(s)) ds \\ &< \mathbf{x}^T(t_0) \mathbf{P} \mathbf{x}(t_0) - b(t - t_0). \end{aligned}$$

The above implies that for a sufficiently large t we would have $\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) < 0$, which contradicts the assumption that $\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) > 0$. This means that $\mathbf{A}$ must be asymptotically stable, and the proof is complete.

◆ Example 4.2

Let

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix},$$

and $\mathbf{Q} = \mathbf{I}_2$. We solve the Lyapunov equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q}$ for $\mathbf{P}$. Assume $\mathbf{P}$ to have the form

$$\mathbf{P} = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix} = \mathbf{P}^T.$$

Then, we have

$$\begin{aligned} \mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} &= \begin{bmatrix} 0 & -1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix} + \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -2p_2 & -p_3 + p_1 - 2p_2 \\ -p_3 + p_1 - 2p_2 & 2p_2 - 4p_3 \end{bmatrix} \\ &= -\mathbf{Q} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}. \end{aligned}$$

Solving the corresponding linear equations yields

$$\mathbf{P} = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}.$$

The solution matrix $\mathbf{P}$ is positive definite. Hence, the matrix $\mathbf{A}$ must have its eigenvalues in the open left-half plane. Indeed, the eigenvalues of $\mathbf{A}$ are both located at -1 .

Definition 4.10 A positive definite function $V(\mathbf{x})$ whose time derivative evaluated on the solutions of the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is negative definite (or negative semidefinite) is called a Lyapunov function for this system.

Lyapunov's theorem gives us a necessary and sufficient condition for asymptotic stability of linear time-invariant dynamical systems. To check if a given matrix $\mathbf{A}$ is asymptotically stable or not, it is enough to take an arbitrary r.s.p.d. matrix $\mathbf{Q}$, solve the equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q}$ for real symmetric $\mathbf{P}$, and check if the obtained $\mathbf{P}$ is positive definite, or not. If $\mathbf{P} > 0$, then $\mathbf{A}$ is asymptotically stable. If $\mathbf{P}$ is not positive definite, then $\mathbf{A}$ cannot be asymptotically stable. Because $\mathbf{Q}$ can be an arbitrary r.s.p.d. matrix, $\mathbf{Q} = \mathbf{I}_n$ (the $n \times n$ identity matrix) is a good choice. It is important to note that there would be no use, as the following example shows, in first choosing a matrix $\mathbf{P}$ and then calculating $\mathbf{Q}$ from the equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q}$.

◆ Example 4.3

Let

$$\mathbf{A} = \begin{bmatrix} -1 & 3 \\ 0 & -1 \end{bmatrix},$$

which is asymptotically stable. Let

$$\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}_2 > 0.$$

Then,

$$\mathbf{Q} = -(\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A}) = \begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix},$$

which is indefinite. Thus, unless $\mathbf{Q}$ turns out to be definite, nothing can be inferred about asymptotic stability from the Lyapunov theorem if we start with $\mathbf{P}$ and then calculate $\mathbf{Q}$.

We now give a geometric interpretation of the Lyapunov theorem. First, note that the function

$$\dot{V} = \frac{dV(\mathbf{x}(t))}{dt}$$

can be expressed, using the chain rule, as

$$\dot{V} = \nabla V^T \dot{\mathbf{x}} = \|\nabla V\| \|\dot{\mathbf{x}}\| \cos \theta,$$

where θ is the angle between the vectors ∇V and $\dot{\mathbf{x}}$. If $\|\dot{\mathbf{x}}\| = 1$, then $\nabla V^T \dot{\mathbf{x}}$ can be viewed as the rate of increase of V at $\mathbf{x}$ in the direction $\dot{\mathbf{x}}$. Thus, $\dot{V} < 0$ if and only if $90^\circ < \theta < 270^\circ$. In such a case the velocity vector $\dot{\mathbf{x}}(t)$ points in the direction of decreasing V . We illustrate the above discussion in Figure 4.5.

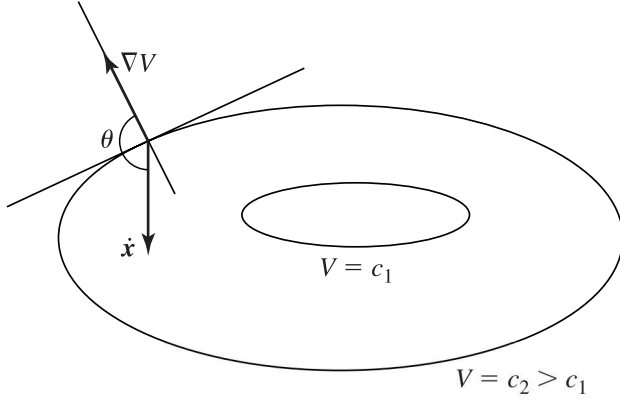


Figure 4.5 Geometric interpretation of the Lyapunov theorem.

We now prove a variant of Theorem 4.1, which is very useful in numerical verification of stability.

Theorem 4.2 A matrix $\mathbf{A}$ is asymptotically stable if and only if for any $\mathbf{Q} = \mathbf{C}^T \mathbf{C}$, $\mathbf{C} \in \mathbb{R}^{p \times n}$, such that the pair $(\mathbf{A}, \mathbf{C})$ is observable, the solution $\mathbf{P}$ to the Lyapunov matrix equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{C}^T \mathbf{C}$ is positive definite.

Proof We first prove the necessity condition ($\Rightarrow$) for asymptotic stability. If $\mathbf{A}$ is asymptotically stable—that is, $\Re \lambda_i(\mathbf{A}) < 0$ —then we will show that

$$\mathbf{P} = \int_0^\infty e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A} t} dt \quad (4.7)$$

is the symmetric positive definite solution to the Lyapunov equation. Indeed, the integrand of (4.7) is a sum of terms of the form

$$t^k e^{\lambda_i t},$$

where λ_i is the i th eigenvalue of $\mathbf{A}$. Since $\Re \lambda_i(\mathbf{A}) < 0$, the integral in (4.7) exists because the terms in $\mathbf{P}$ do not diverge when $t \rightarrow \infty$. Indeed, if we let $-\alpha_i = \Re \lambda_i$, where $\alpha_i > 0$, then

$$\lim_{t \rightarrow \infty} t^k e^{-\alpha_i t} = \lim_{t \rightarrow \infty} \frac{t^k}{e^{\alpha_i t}},$$

We differentiate the numerator and denominator of the above k times and use L'Hôpital's rule to get

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{t^k}{e^{\alpha_i t}} &= \lim_{t \rightarrow \infty} \frac{k t^{k-1}}{\alpha_i e^{\alpha_i t}} \\ &= \lim_{t \rightarrow \infty} \frac{k(k-1) t^{k-2}}{\alpha_i^2 e^{\alpha_i t}} \\ &\vdots \\ &= \lim_{t \rightarrow \infty} \frac{k!}{\alpha_i^k e^{\alpha_i t}} \\ &= 0. \end{aligned} \quad (4.8)$$

Thus, by (4.8), the formula for $\mathbf{P}$ given by (4.7) is well-defined. Next, note that $\mathbf{P} = \mathbf{P}^T$. The matrix $\mathbf{P}$ is also positive definite because the pair $(\mathbf{A}, \mathbf{C})$ is observable. Substituting the expression for $\mathbf{P}$ into the Lyapunov equation and performing manipulations yields

$$\begin{aligned} \mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} &= \int_0^\infty \mathbf{A}^T e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A} t} dt + \int_0^\infty e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A} t} \mathbf{A} dt \\ &= \int_0^\infty \frac{d}{dt} (e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A} t}) dt \\ &= e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A} t} \Big|_0^\infty \\ &= -\mathbf{C}^T \mathbf{C}. \end{aligned}$$

The sufficiency part ($\Leftarrow$) can be proven using the arguments of the proof of the sufficiency part of the Lyapunov theorem and noting that the observability of the pair $(\mathbf{A}, \mathbf{C})$ implies that $\mathbf{x}^T(t) \mathbf{C}^T \mathbf{C} \mathbf{x}(t)$ is not identically zero along any nonzero solution of $\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t)$.

We now show that if $\mathbf{A}$ is asymptotically stable, then the Lyapunov equation (4.6) has a unique solution for every $\mathbf{Q}$. Uniqueness can be proven in many ways. We use the Kronecker product in our proof. Using the results of Section A.5, we represent the Lyapunov equation (4.6) as

$$(\mathbf{I}_n \otimes \mathbf{A}^T + \mathbf{A}^T \otimes \mathbf{I}_n) s(\mathbf{P}) = -s(\mathbf{Q}). \quad (4.9)$$

The condition for a solution $s(\mathbf{P})$ of the above equation to be unique is

$$\lambda_i(\mathbf{A}^T) + \lambda_j(\mathbf{A}) \neq 0, \quad i, j = 1, 2, \dots, n.$$

Because $\lambda_i(\mathbf{A}^T) = \lambda_i(\mathbf{A})$, and $\mathbf{A}$ is asymptotically stable by assumption, the condition $\lambda_i(\mathbf{A}^T) + \lambda_j(\mathbf{A}) \neq 0$ is met, and hence the Lyapunov equation has a unique solution.

4.4 Evaluating Quadratic Indices

In this section we apply the Lyapunov theory to evaluate the quadratic indices

$$J_r = \int_0^\infty t^r \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) dt, \quad r = 0, 1, \dots$$

subject to

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t), \quad \mathbf{x}(0) = \mathbf{x}_0,$$

where $\mathbf{Q} = \mathbf{C}^T \mathbf{C}$ such that the pair $(\mathbf{A}, \mathbf{C})$ is observable, and the matrix $\mathbf{A}$ is asymptotically stable. We can reformulate the above problem as follows. Evaluate the quadratic performance indices

$$J_r = \int_0^\infty t^r \mathbf{y}^T(t) \mathbf{y}(t) dt, \quad r = 0, 1, \dots$$

subject to

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t), & \mathbf{x}(0) &= \mathbf{x}_0, \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t),\end{aligned}$$

where the pair $(\mathbf{A}, \mathbf{C})$ is observable, and the matrix $\mathbf{A}$ is asymptotically stable.

To evaluate

$$J_0 = \int_0^\infty \mathbf{y}^T(t) \mathbf{y}(t) dt = \int_0^\infty \mathbf{x}^T(t) \mathbf{C}^T \mathbf{C} \mathbf{x}(t) dt = \int_0^\infty \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) dt,$$

we note that

$$\frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) = -\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t), \quad (4.10)$$

where $\mathbf{P}$ and $\mathbf{Q} = \mathbf{C}^T \mathbf{C}$ satisfy the Lyapunov matrix equation, $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q}$. Hence,

$$J_0 = - \int_0^\infty \frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) dt. \quad (4.11)$$

Integrating both sides of (4.10) with respect to t yields

$$\begin{aligned}J_0 &= - \int_0^\infty \frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) dt \\ &= - (\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) \Big|_0^\infty \\ &= \lim_{t \rightarrow \infty} (-\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) + \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0) \\ &= \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0)\end{aligned}$$

because, by assumption, $\mathbf{A}$ is asymptotically stable and therefore $\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{0}$ for all $\mathbf{x}(0)$. Furthermore, $\mathbf{P} = \mathbf{P}^T > 0$, and thus $J_0 > 0$ for all $\mathbf{x}(0) \neq \mathbf{0}$.

Repeating the above arguments leads to a similar expression for J_1 . First observe that

$$\begin{aligned}\frac{d}{dt}(t \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) &= \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) + t \frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) \\ &= \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) - t \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t),\end{aligned}$$

where $\mathbf{P}$ satisfies the Lyapunov matrix equation, $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q}$. Hence,

$$\begin{aligned}J_1 &= \int_0^\infty t \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) dt \\ &= \int_0^\infty \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) dt - \int_0^\infty \frac{d}{dt}(t \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) dt.\end{aligned} \quad (4.12)$$

Because $\mathbf{P} = \mathbf{P}^T > 0$, we can solve

$$\mathbf{A}^T \mathbf{P}_1 + \mathbf{P}_1 \mathbf{A} = -\mathbf{P}$$

for $\mathbf{P}_1 = \mathbf{P}_1^T > 0$. The above equation can also be represented as

$$\frac{d}{dt}(\mathbf{x}^T(t)\mathbf{P}_1\mathbf{x}(t)) = -\mathbf{x}^T(t)\mathbf{P}\mathbf{x}(t). \quad (4.13)$$

Substituting (4.13) into (4.12) yields

$$\begin{aligned} J_1 &= - \int_0^\infty \frac{d}{dt}(\mathbf{x}^T(t)\mathbf{P}_1\mathbf{x}(t))dt - (\mathbf{x}^T(t)\mathbf{P}\mathbf{x}(t)) \Big|_0^\infty \\ &= \lim_{t \rightarrow \infty} (-\mathbf{x}^T(t)\mathbf{P}_1\mathbf{x}(t)) + \mathbf{x}^T(0)\mathbf{P}_1\mathbf{x}(0) - 0 + 0 \\ &= \mathbf{x}^T(0)\mathbf{P}_1\mathbf{x}(0). \end{aligned}$$

Using similar arguments, we can evaluate J_r , $r > 1$, which involves successive solutions to the Lyapunov matrix equations.

In many control problems a dynamical system model is given in terms of a transfer function rather than in terms of a system of differential equations. We now discuss a method of converting a given rational function into a system of first-order differential equations. After this process is completed, we use the Lyapunov theory to evaluate a performance index of interest. Suppose we are given a rational function of the form

$$E(s) = \frac{b_{n-1}s^{n-1} + \cdots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0},$$

where $E(s) = \mathcal{L}(e(t))$ is the Laplace transform of $e(t)$. We now show that $e(t)$ can be generated as the solution of the differential equation

$$\frac{d^n e(t)}{dt^n} + a_{n-1} \frac{d^{n-1} e(t)}{dt^{n-1}} + \cdots + a_1 \frac{de(t)}{dt} + a_0 e(t) = 0, \quad (4.14)$$

subject to a set of initial conditions that we shall derive. We follow the method of Power and Simpson [237, pp. 220–221]. We first take the Laplace transform of the differential equation and then rearrange the resulting terms in an appropriate matrix equation. Recall that

$$\mathcal{L}\left(\frac{d^i e(t)}{dt^i}\right) = s^i E(s) - s^{i-1}e(0) - s^{i-2} \frac{de(0)}{dt} - \cdots - \frac{d^{i-1}e(0)}{dt^{i-1}}. \quad (4.15)$$

Applying the above to (4.14) and comparing corresponding terms with $E(s)$ leads to the equation for the initial conditions:

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_{n-1} & 1 \\ a_2 & a_3 & \cdots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n-1} & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} e(0) \\ \frac{de(0)}{dt} \\ \vdots \\ \frac{d^{n-2}e(0)}{dt^{n-2}} \\ \frac{d^{n-1}e(0)}{dt^{n-1}} \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix}.$$

The coefficient matrix in the above equation is nonsingular. Thus, there is a unique solution to this equation, that is, we can uniquely determine the initial conditions. We next define the state vector

$$\mathbf{x} = \left[e \quad \frac{de}{dt} \quad \cdots \quad \frac{d^{n-1}e}{dt^{n-1}} \right]^T,$$

and represent the differential equation (4.14) as

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x},$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}.$$

The vector of initial conditions is

$$\mathbf{x}(0) = \begin{bmatrix} e(0) \\ \frac{de(0)}{dt} \\ \vdots \\ \frac{d^{n-2}e(0)}{dt^{n-2}} \\ \frac{d^{n-1}e(0)}{dt^{n-1}} \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & \cdots & a_{n-1} & 1 \\ a_2 & a_3 & \cdots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n-1} & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix}.$$

Note that $e = [1 \ 0 \ \cdots \ 0 \ 0]\mathbf{x}$.

The above procedure for converting a rational function description of a process into its equivalent state-space representation shows a relation between the zeros of the given rational function and the initial conditions of the associated state-space realization.

We will now use the above method of constructing a state-space realization of a given rational function in the following example that involves evaluation of a quadratic index.

◆ Example 4.4

Consider a block diagram of a dynamical control system model shown in Figure 4.6. This system comes from Dorf and Bishop [66, p. 257]. The values of the parameters are $K_1 = 0.5$ and $K_1 K_2 K_p = 2.5$. We shall use the Lyapunov theory to select the gain K_3 that minimizes the effect of the disturbance $u(t) = \mathcal{L}^{-1}(U(s))$ in the sense of the integral of the square error (ISE) criterion. In other words, we wish to find K_3 that leads to minimization of the performance index

$$J_0 = \int_0^\infty c^2(t) dt,$$

where $c(t) = \mathcal{L}^{-1}(C(s))$. We assume a unit step disturbance and $R(s) = 0$.

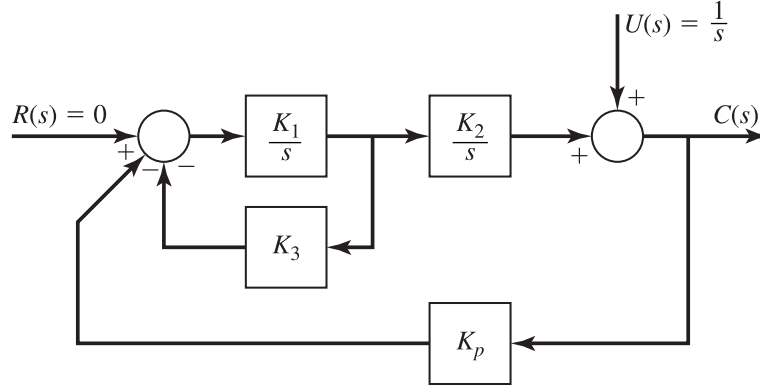


Figure 4.6 Block diagram of the optimized control system model of Example 4.4.

We first find a rational function that models $C(s)$. From the block diagram shown in Figure 4.6, we obtain

$$C(s) = U(s) - \frac{K_2}{s} \frac{K_1}{s + K_1 K_3} K_p C(s).$$

Hence

$$C(s) = \frac{s + 0.5 K_3}{s^2 + 0.5 K_3 s + 2.5}.$$

The next step is to determine the state-space representation corresponding to $C(s)$. We use the method given before this example. We first form the differential equation for $c(t)$:

$$\frac{d^2 c(t)}{dt^2} + 0.5 K_3 \frac{dc(t)}{dt} + 2.5 c(t) = 0$$

with initial conditions

$$\begin{bmatrix} 0.5 K_3 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c(0) \\ \frac{dc(0)}{dt} \end{bmatrix} = \begin{bmatrix} 0.5 K_3 \\ 1 \end{bmatrix}.$$

Solving the above equation yields

$$\begin{bmatrix} c(0) \\ \frac{dc(0)}{dt} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Let $x_1 = c$, $x_2 = \frac{dc}{dt}$. The differential equation in vector-matrix form corresponding to $C(s)$ is

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 \\ -2.5 & -0.5 K_3 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$c(t) = \mathbf{c} \mathbf{x}(t),$$

where $\mathbf{c} = [1 \ 0]$. We are now ready to evaluate the performance index J_0 :

$$J_0 = \int_0^\infty c^2(t) dt = \int_0^\infty \mathbf{x}^T(t) \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}(t) dt = \int_0^\infty \mathbf{x}^T(t) \mathbf{c}^T \mathbf{c} \mathbf{x}(t) dt.$$

The pair $(\mathbf{A}, \mathbf{c})$ is observable. Therefore, we can solve the Lyapunov equation

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{c}^T \mathbf{c}$$

for $\mathbf{P}$ to obtain

$$\mathbf{P} = \begin{bmatrix} \frac{K_3}{10} + \frac{1}{K_3} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5K_3} \end{bmatrix}.$$

Hence

$$J_0 = \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0) = \frac{K_3}{10} + \frac{1}{K_3}.$$

Differentiating J_0 and equating the result to zero gives

$$\frac{dJ_0}{dK_3} = \frac{1}{10} - \frac{1}{K_3^2} = 0.$$

Selecting

$$K_3 = \sqrt{10}$$

yields a positive definite $\mathbf{P}$. Note that because

$$\left. \frac{d^2 J_0}{dK_3^2} \right|_{K_3=\sqrt{10}} = \left. \frac{2}{K_3^3} \right|_{K_3=\sqrt{10}} > 0,$$

that is, the second-order sufficiency condition for the minimum is satisfied, we conclude that $K_3 = \sqrt{10} \approx 3.16$ minimizes J_0 .

We now wish to determine the gain K_3 that leads to minimization of the effect of the disturbance $u(t)$ in the sense of the integral of time multiplied by the squared error (ITSE) criterion. That is, we wish to select K_3 to minimize the performance index:

$$J_1 = \int_0^\infty t c^2(t) dt.$$

We first solve the Lyapunov equation,

$$\mathbf{A}^T \mathbf{P}_1 + \mathbf{P}_1 \mathbf{A} = -\mathbf{P},$$

to get

$$\mathbf{P}_1 = \begin{bmatrix} \frac{2}{K_3^2} + \frac{K_3^2}{100} & \frac{1}{5} \left(\frac{1}{K_3} + \frac{K_3}{10} \right) \\ \frac{1}{5} \left(\frac{1}{K_3} + \frac{K_3}{10} \right) & \frac{4}{5K_3^2} + \frac{1}{25} \end{bmatrix}.$$

Hence

$$J_1 = \mathbf{x}^T(0) \mathbf{P}_1 \mathbf{x}(0) = \frac{2}{K_3^2} + \frac{K_3^2}{100}.$$

Differentiating J_1 and equating the result to zero gives

$$\frac{dJ_1}{dK_3} = -\frac{4}{K_3^3} - \frac{K_3}{50} = 0.$$

We select $K_3 = \sqrt[4]{200}$ to keep $\mathbf{P}_1$ positive definite. For this choice of K_3 the second-order sufficiency condition for the minimum of J_1 is satisfied. Thus, $K_3 \approx 3.76$ minimizes J_1 .

4.5 Discrete-Time Lyapunov Equation

We will now use the Lyapunov approach to the stability analysis of the discrete-time linear time-invariant systems modeled by

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k), \quad \mathbf{x}(k_0) = \mathbf{x}_0. \quad (4.16)$$

By analogy with the continuous-time case, consider a quadratic positive definite form,

$$V(\mathbf{x}(k)) = \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k).$$

We evaluate the first difference of V , defined as $\Delta V = V(\mathbf{x}(k+1)) - V(\mathbf{x}(k))$, on the trajectories of (4.16) to obtain

$$\begin{aligned} \Delta V(\mathbf{x}(k)) &= V(\mathbf{x}(k+1)) - V(\mathbf{x}(k)) \\ &= \mathbf{x}^T(k+1) \mathbf{P} \mathbf{x}(k+1) - \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k) \\ &= \mathbf{x}^T(k) \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{x}(k) - \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k) \\ &= \mathbf{x}^T(k) (\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P}) \mathbf{x}(k), \end{aligned} \quad (4.17)$$

where we replaced $\mathbf{x}(k+1)$ in the above with $\mathbf{A}\mathbf{x}(k)$. For the solutions of (4.16) to decrease the “energy function” V , at each $k > k_0$, it is necessary and sufficient that $\Delta V(\mathbf{x}(k)) < 0$, or, equivalently, that for some positive definite $\mathbf{Q}$,

$$\Delta V(\mathbf{x}(k)) = -\mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k). \quad (4.18)$$

Combining (4.16) and (4.18), we obtain the so-called *discrete-time Lyapunov equation*,

$$\boxed{\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P} = -\mathbf{Q}.} \quad (4.19)$$

Theorem 4.3 The matrix $\mathbf{A}$ has its eigenvalues in the open unit disk if and only if for any positive definite $\mathbf{Q}$ the solution $\mathbf{P}$ to (4.19) is positive definite.

Proof ($\Rightarrow$) We can use the Kronecker product, and in particular relation (A.62), to represent the discrete-time Lyapunov equation as

$$(\mathbf{I}_{n^2} - \mathbf{A}^T \otimes \mathbf{A}^T) \mathbf{s}(\mathbf{P}) = \mathbf{s}(\mathbf{Q}), \quad (4.20)$$

where $\otimes$ denotes the Kronecker product and $\mathbf{s}(\cdot)$ is the stacking operator. It is easy to see that the above equation has a unique solution if and only if

$$1 - \lambda_i \lambda_j \neq 0 \quad \text{for all } i, j = 1, 2, \dots, n,$$

where $\{\lambda_i\}$ is the set of eigenvalues of $\mathbf{A}$. By assumption, $|\lambda_i| < 1$, which is a sufficient condition for (4.20) and hence (4.19) to have a unique solution. The solution $\mathbf{P}$ can be expressed as

$$\mathbf{P} = \sum_{k=0}^{\infty} (\mathbf{A}^T)^k \mathbf{Q} \mathbf{A}^k. \quad (4.21)$$

The above expression is well-defined because all eigenvalues of $\mathbf{A}$ have magnitude (strictly) less than unity. Substituting (4.21) into (4.19) gives

$$\begin{aligned} \mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P} &= \mathbf{A}^T \left(\sum_{k=0}^{\infty} (\mathbf{A}^T)^k \mathbf{Q} \mathbf{A}^k \right) \mathbf{A} - \sum_{k=0}^{\infty} (\mathbf{A}^T)^k \mathbf{Q} \mathbf{A}^k \\ &= \sum_{k=1}^{\infty} (\mathbf{A}^T)^k \mathbf{Q} \mathbf{A}^k - \sum_{k=1}^{\infty} (\mathbf{A}^T)^k \mathbf{Q} \mathbf{A}^k - \mathbf{Q} \\ &= -\mathbf{Q}. \end{aligned}$$

Thus, indeed (4.21) satisfies the discrete-time Lyapunov equation. Note that if $\mathbf{Q}$ is symmetric, so is $\mathbf{P}$. If $\mathbf{Q}$ is positive definite, so is $\mathbf{P}$.

($\Leftarrow$) Let λ_i be an eigenvalue of $\mathbf{A}$ and let $\mathbf{v}_i$ be a corresponding eigenvector; that is, $\mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i$. Observe that λ_i and $\mathbf{v}_i$ may be complex. We denote by $\mathbf{v}_i^*$ the complex conjugate transpose of $\mathbf{v}_i$. Note that $\mathbf{v}_i^* \mathbf{P} \mathbf{v}_i$ is a scalar. We will show that its complex conjugate equals itself, which means that $\mathbf{v}_i^* \mathbf{P} \mathbf{v}_i$ is real. Indeed,

$$\overline{\mathbf{v}_i^* \mathbf{P} \mathbf{v}_i} = \mathbf{v}_i^* \mathbf{P}^* \mathbf{v}_i = \mathbf{v}_i^* \mathbf{P}^T \mathbf{v}_i = \mathbf{v}_i^* \mathbf{P} \mathbf{v}_i,$$

where we used the fact that the complex conjugate transpose of a real matrix reduces to its transpose. Thus, $\mathbf{v}_i^* \mathbf{P} \mathbf{v}_i$ is real for any complex $\mathbf{v}_i$.

Premultiplying now (4.19) by $\mathbf{v}_i^*$ and postmultiplying by $\mathbf{v}_i$ gives

$$\mathbf{v}_i^* (\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P}) \mathbf{v}_i = -\mathbf{v}_i^* \mathbf{Q} \mathbf{v}_i. \quad (4.22)$$

The left-hand side of the above equality evaluates to

$$\begin{aligned} \mathbf{v}_i^* \mathbf{A}^T \mathbf{P} \mathbf{A} \mathbf{v}_i - \mathbf{v}_i^* \mathbf{P} \mathbf{v}_i &= \bar{\lambda}_i \mathbf{v}_i^* \mathbf{P} \mathbf{v}_i \lambda_i - \mathbf{v}_i^* \mathbf{P} \mathbf{v}_i \\ &= (|\lambda_i|^2 - 1) \mathbf{v}_i^* \mathbf{P} \mathbf{v}_i. \end{aligned}$$

Comparing the above with the right-hand side of (4.22) yields

$$(1 - |\lambda_i|^2) \mathbf{v}_i^* \mathbf{P} \mathbf{v}_i = \mathbf{v}_i^* \mathbf{Q} \mathbf{v}_i > 0$$

because $\mathbf{Q}$ is positive definite. Because $\mathbf{P}$ is positive definite, we have to have $|\lambda_i| < 1$. The proof is complete.

We will now concern ourselves with evaluating performance indices on the trajectories of (4.16). Specifically, we wish to evaluate

$$K_r = \sum_{k=0}^{\infty} k^r \mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k), \quad r = 0, 1, \dots \quad (4.23)$$

where $\mathbf{Q} = \mathbf{Q}^T > 0$ subject to (4.16). We assume that the matrix $\mathbf{A}$ has its eigenvalues in the open unit disk; that is, the matrix $\mathbf{A}$ is *convergent*. We first consider the case when $k = 0$. For this case, K_r becomes

$$K_0 = \sum_{k=0}^{\infty} \mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k). \quad (4.24)$$

Because the matrix $\mathbf{A}$ is convergent, the solution $\mathbf{P}$ to (4.19) is positive definite. Using (4.17), (4.18), and (4.16), we write

$$\begin{aligned} \mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k) &= -\mathbf{x}^T(k) (\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P}) \mathbf{x}(k) \\ &= \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k) - \mathbf{x}^T(k+1) \mathbf{P} \mathbf{x}(k+1). \end{aligned} \quad (4.25)$$

Summing both sides of the above from $k = 0$ to $k = \infty$ and using the fact that $\mathbf{x}(\infty) = \mathbf{0}$ because the matrix $\mathbf{A}$ is convergent, we obtain

$$K_0 = \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0). \quad (4.26)$$

We will now evaluate

$$K_1 = \sum_{k=0}^{\infty} k \mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k),$$

subject to (4.16). To proceed, we multiply both sides of (4.25) by k to get

$$k \mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k) = k \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k) - k \mathbf{x}^T(k+1) \mathbf{P} \mathbf{x}(k+1).$$

Summing the above from $k = 0$ to $k = \infty$ gives

$$\begin{aligned} K_1 &= \mathbf{x}^T(1) \mathbf{P} \mathbf{x}(1) - \mathbf{x}^T(2) \mathbf{P} \mathbf{x}(2) + 2\mathbf{x}^T(2) \mathbf{P} \mathbf{x}(2) - 2\mathbf{x}^T(3) \mathbf{P} \mathbf{x}(3) + \dots \\ &= \sum_{k=0}^{\infty} \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k) - \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0). \end{aligned} \quad (4.27)$$

Using the method we employed to find the value of K_0 , we now evaluate the first term on the right-hand side of (4.27) to get

$$\sum_{k=0}^{\infty} \mathbf{x}^T(k) \mathbf{P} \mathbf{x}(k) = \mathbf{x}^T(0) \mathbf{P}_1 \mathbf{x}(0),$$

where $\mathbf{P}_1$ satisfies

$$\mathbf{A}^T \mathbf{P}_1 \mathbf{A} - \mathbf{P}_1 = -\mathbf{P}.$$

Taking the above into account, we represent (4.27) as

$$K_1 = \mathbf{x}^T(0) (\mathbf{P}_1 - \mathbf{P}) \mathbf{x}(0).$$

Proceeding in a similar manner, we can find expressions for K_r , where $r > 1$.

4.6 Constructing Robust Linear Controllers

In this section we use the Lyapunov theorem to construct robust state-feedback controllers. One of the basic issues in the control of dynamical systems is the effect of uncertainties, or neglected nonlinearities, on the stability of the closed-loop system. A controller is viewed as robust if it maintains stability for all uncertainties in an expected range. We consider a class of dynamical systems modeled by

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{f}(t, \mathbf{x}(t)) + \mathbf{B}\mathbf{h}(t, \mathbf{x}(t), \mathbf{u}(t)), \quad (4.28)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$, $\mathbf{u}(t) \in \mathbb{R}^m$, and the functions $\mathbf{h}$ and $\mathbf{f}$ model uncertainties, or nonlinearities, in the system. We refer to $\mathbf{h}$ as the *matched uncertainty* because it affects the system dynamics via the input matrix $\mathbf{B}$ in the same fashion as the input $\mathbf{u}$ does. In other words, the uncertainty $\mathbf{h}$ matches the system input $\mathbf{u}$. The vector $\mathbf{f}$ models the *unmatched uncertainty*. We assume that the uncertain elements $\mathbf{h}$ and $\mathbf{f}$ satisfy the following norm bounds:

1. $\|\mathbf{h}(t, \mathbf{x}, \mathbf{u})\| \leq \gamma_h \|\mathbf{u}\| + \alpha_h \|\mathbf{x}\|$,
2. $\|\mathbf{f}(t, \mathbf{x})\| \leq \alpha_f \|\mathbf{x}\|$,

where γ_h , α_h , and α_f are known nonnegative constants. We further assume that the matrix $\mathbf{A}$ is asymptotically stable. If this is not the case, we apply a preliminary state-feedback controller $\mathbf{u} = -\mathbf{K}\mathbf{x}$ such that $\mathbf{A} - \mathbf{B}\mathbf{K}$ is asymptotically stable. Such a feedback exists provided that the pair $(\mathbf{A}, \mathbf{B})$ is stabilizable. Our goal is to construct a linear state-feedback controller that would make the closed-loop system asymptotically stable for arbitrary $\mathbf{f}$ and $\mathbf{h}$ that satisfy the above norm bounds.

Theorem 4.4 Suppose that $\mathbf{A}$ is asymptotically stable and that $\mathbf{P} = \mathbf{P}^T > 0$ is the solution to the Lyapunov matrix equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -2\mathbf{Q}$ for some $\mathbf{Q} = \mathbf{Q}^T > 0$. Suppose also that

$$\alpha_f < \frac{\lambda_{\min}(\mathbf{Q})}{\lambda_{\max}(\mathbf{P})} \quad \text{and} \quad \gamma_h < 1.$$

Then, the state-feedback controller $\mathbf{u} = -\gamma \mathbf{B}^T \mathbf{P} \mathbf{x}$, where

$$\gamma > \frac{\alpha_h^2}{4(\lambda_{\min}(\mathbf{Q}) - \alpha_f \lambda_{\max}(\mathbf{P}))(1 - \gamma_h)}$$

stabilizes the uncertain system model (4.28) for arbitrary $\mathbf{f}$ and $\mathbf{h}$ that satisfy the norm bounds.

Proof The time derivative of the positive definite function $V = \mathbf{x}^T \mathbf{P} \mathbf{x}$ evaluated on any trajectory of the closed-loop system is

$$\begin{aligned} \dot{V} &= 2\mathbf{x}^T \mathbf{P} \dot{\mathbf{x}} \\ &= -2\mathbf{x}^T \mathbf{Q} \mathbf{x} - 2\gamma \mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{B}^T \mathbf{P} \mathbf{x} + 2\mathbf{x}^T \mathbf{P} \mathbf{f} + 2\mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{h}. \end{aligned}$$

Our goal is to determine $\tilde{\gamma} > 0$ such that if $\gamma > \tilde{\gamma}$, then $\dot{V} < 0$, which in turn implies the asymptotic stability of the closed-loop system. To proceed further, recall that if $\mathbf{Q}$ is a symmetric matrix, then

$$\lambda_{\min}(\mathbf{Q}) \|\mathbf{x}\|^2 \leq \mathbf{x}^T \mathbf{Q} \mathbf{x} \leq \lambda_{\max}(\mathbf{Q}) \|\mathbf{x}\|^2,$$

and therefore

$$-\mathbf{x}^T \mathbf{Q} \mathbf{x} \leq -\lambda_{\min}(\mathbf{Q}) \|\mathbf{x}\|^2.$$

Furthermore,

$$\mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{B}^T \mathbf{P} \mathbf{x} = \mathbf{x}^T \mathbf{P} \mathbf{B} (\mathbf{x}^T \mathbf{P} \mathbf{B})^T = \|\mathbf{x}^T \mathbf{P} \mathbf{B}\|^2.$$

For a symmetric positive definite matrix $\mathbf{P}$, its induced 2-norm is

$$\|\mathbf{P}\| = \lambda_{\max}(\mathbf{P}).$$

Taking the above relations into account, using the norm bounds on the uncertain elements, and performing some manipulations yields

$$\begin{aligned} \dot{V} \leq & -2\lambda_{\min}(\mathbf{Q}) \|\mathbf{x}\|^2 - 2\gamma \|\mathbf{x}^T \mathbf{P} \mathbf{B}\|^2 + 2\alpha_f \lambda_{\max}(\mathbf{P}) \|\mathbf{x}\|^2 \\ & + 2\|\mathbf{x}^T \mathbf{P} \mathbf{B}\|(\alpha_h \|\mathbf{x}\| + \gamma_h \gamma \|\mathbf{x}^T \mathbf{P} \mathbf{B}\|). \end{aligned} \quad (4.29)$$

Let

$$\tilde{\mathbf{Q}} = \begin{bmatrix} \lambda_{\min}(\mathbf{Q}) - \alpha_f \lambda_{\max}(\mathbf{P}) & -\frac{\alpha_h}{2} \\ -\frac{\alpha_h}{2} & \gamma(1 - \gamma_h) \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \quad (4.30)$$

Then, we can represent (4.29) as

$$\dot{V} \leq -2 \begin{bmatrix} \|\mathbf{x}\| & \|\mathbf{x}^T \mathbf{P} \mathbf{B}\| \end{bmatrix} \tilde{\mathbf{Q}} \begin{bmatrix} \|\mathbf{x}\| \\ \|\mathbf{x}^T \mathbf{P} \mathbf{B}\| \end{bmatrix}. \quad (4.31)$$

By Theorem A.6 on page 634, for $\dot{V}$ to be negative it is enough that the leading principal minors of the matrix $\tilde{\mathbf{Q}}$ given by (4.30) be positive. The first-order leading principal minor is positive by assumption. For the second-order leading principal minor to be positive it is sufficient that $\gamma_h < 1$ and

$$\gamma > \frac{\alpha_h^2}{4(\lambda_{\min}(\mathbf{Q}) - \alpha_f \lambda_{\max}(\mathbf{P}))(1 - \gamma_h)}.$$

The proof is now complete.

Note that if $\gamma_h = 0$, then the closed-loop system is asymptotically stable if

$$\alpha_f < \frac{\lambda_{\min}(\mathbf{Q})}{\lambda_{\max}(\mathbf{P})} \quad \text{and} \quad \gamma > \frac{\alpha_h^2}{4(\lambda_{\min}(\mathbf{Q}) - \alpha_f \lambda_{\max}(\mathbf{P}))}.$$